

A Calibrated Multi-Salt Radiolysis Model for Molten Chloride and Fluoride Reactor Fuels: Application of Hierarchical Bayesian Mechanism-Adequacy Estimation to Operational MSR Predictions

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Abstract

The molten salt reactor (MSR) concept requires quantitative predictions of how the salt’s chemical inventory evolves under sustained radiation, on timescales spanning nanoseconds through decades. We construct a calibrated multi-salt radiolysis model for chloride and fluoride MSR fuels by combining a mechanistic reaction network with a digitized corpus of pulse-radiolysis and chronic gamma-irradiation literature. The likelihood-bearing calibration corpus contains 73 observations drawn from 13 of the 14 digitized source papers in Table 2. The model is calibrated through the Hierarchical Bayesian Mechanism-Adequacy Estimation (HBMAE) framework of our companion methodology paper (Tanore-Tamales, 2026), summarized in Section 1.1. The integrated 24-parameter posterior recovers the published Iwamatsu Arrhenius parameters for chromium and zinc within 1σ and quantifies the Pikaev–Iwamatsu inter-facility offset at $b^{(\text{Pikaev})} = -3.35 [-4.53, -2.44]$. A meta-hierarchical chemistry-feature regression on charge, ionic radius, and electronegativity recovers in-corpus microsecond-Pikaev metals within ~ 1 log unit but under-predicts the LEAF-picosecond corpus by 2–4 log units; the regression is therefore descriptive rather than predictive for fission-product elements whose operational chemistry is not yet represented at any facility in the training set. We deploy the calibrated model to two operational MSR design problems. A 500 MW(th) NaCl–UCl₃ fast-spectrum reactor yields, under the Phillips-bounded U(III)/U(IV) kernel, a posterior on cover-gas P_{Cl_2} whose median lies far below a notional 100 Pa screening threshold; this margin is dominated by the assumed U-redox sink and would collapse if the U-buffer is incomplete (§8.1.1). A FLiBe–UF₄ salt loop predicts a steady-state cover-gas P_{F_2} of 31 Pa (median; 90 % CI [19, 44] Pa), within a factor of three of the same screening threshold and dominated by the Davis-2022 $G(\text{F}_2)$ and the Toth–Felter recombination E_a extrapolation from 423 K to 873 K. Posterior variance attribution and three recommended experiments are reported in Section 8.

Keywords: molten salt reactor; radiolysis; chloride salt; fluoride salt; NaCl–UCl₃; FLiBe; chemical kinetic network; Bayesian calibration; uncertainty quantification.

1 Introduction

The molten salt reactor concept has re-emerged in the past decade as a leading candidate for advanced fission heat generation, with active commercial development of fast-spectrum chloride-

fueled designs (Phillips et al., 2022; TerraPower, 2021) and thermal-spectrum fluoride-cooled and fluoride-fueled designs (Forsberg, 2020; Andreades et al., 2016; Holcomb and Cetiner, 2010). Each of these designs rests on a quantitative understanding of the chemical evolution of the salt under sustained ionizing radiation. At time scales ranging from nanoseconds (solvated electrons and primary radical recombination) through years (cumulative redox drift and accumulation of gaseous radiolysis products in the cover-gas system), the salt chemistry governs four engineering outcomes simultaneously:

1. the *redox state* of the salt, which determines structural corrosion rates against Hastelloy, SS-316 and Mo-Re alloys (Zhang et al., 2018; Olander, 2002);
2. the *cover-gas inventory* of molecular Cl_2 in chloride systems or F_2 in fluoride systems, which governs off-gas system design and accident-source-term calculations (Sohal et al., 2010; International Atomic Energy Agency, 2013);
3. the *speciation of dissolved fission products*, in particular the partitioning of iodine, tellurium, and cesium between salt and cover gas, which determines containment release pathways (Briggs, 1964; Williams et al., 2006);
4. the *long-term thermophysical stability* of the salt itself, which determines fuel re-processing and salt cleanup requirements (Romatoski and Hu, 2017).

The pulse-radiolysis literature that supplies the kinetic underpinnings of this chemistry has grown substantially over the last five years. The Brookhaven LEAF program has produced quantitative kinetics for the solvated electron e_s^- and the dichlorine radical anion $\text{Cl}_2^{\bullet-}$ in molten LiCl-KCl eutectic at 400°C to 600°C (Iwamatsu et al., 2022), for chromium-mediated electron transfer in the same melt (Iwamatsu et al., 2026), for iodide perturbation of the chloride radical pool (Conrad et al., 2023), for neodymium speciation as a representative lanthanide (Castro Baldivieso et al., 2026), and for the first actinide rate constant in chloride media (Cf^{3+} oxidation by $\text{Cl}_2^{\bullet-}$ and $\text{SO}_4^{\bullet-}$) (Rotermund et al., 2024). Soviet-era pulse-radiolysis measurements (Pikaev et al., 1982; Makarov et al., 1982; Hagiwara et al., 1987) established the basic transient inventory across the alkali-halide series at the order-of-magnitude level. Ab initio molecular dynamics calculations (Kristoffersen and Metiu, 2018) provide thermodynamic-feasibility anchors for the chloride redox sub-network. Steady-state fluorine-gas yields from gamma-irradiated LiF , BeF_2 , UF_4 , ThF_4 , and FLiBe powders have been recently quantified (Davis et al., 2022), complementing the early Toth-Felker recombination measurements (Toth and Felker, 1990). In the chloride-salt nuclear fuel context, Phillips and coworkers (Phillips et al., 2022) reported the first long-duration gamma-irradiation of NaCl-UCl_3 salt with no detectable Cl_2 above approximately 1000 ppm after 31 MGy of dose, providing a critical censored upper bound for calibration.

Despite this experimental progress, no published compilation simultaneously calibrates against the chloride and fluoride corpora in a single framework within which an operational MSR designer can run an auditable, uncertainty-quantified cover-gas Cl_2 or F_2 prediction. The available kinetic compilations are either single-paper Arrhenius fits with no cross-host transferability, hand-tuned multi-paper averages without formal calibration, or theoretical/AIMD computations that have not been validated against an explicitly digitized multi-modality corpus; Mausolf and coworkers (Phillips et al., 2022) review the closest related chloride compilation but do not perform a full multi-modality Bayesian calibration. This paper closes that gap.

The methodological backbone of the calibration is a hierarchical Bayesian framework that jointly handles transient time-series, scalar rate constants, derived Arrhenius pairs, one-sided censored

detection limits, G-values, and molar absorptivities, while resolving inter-facility systematic biases and cross-host chemistry-feature transfer. The full development is in our companion methodology paper (Tanore-Tamales, 2026); we refer to that paper for the formal theorems on identifiability and contraction and for the synthetic-benchmark comparison against four state-of-the-art comparator methods. Section 1.1 below summarizes the five-layer construction sufficiently to read the present paper without consulting Paper I. The present paper is the *application* companion: we take the calibrated 24-parameter HBMAE posterior as given, extend it with a chemistry-feature regression that supplies predictions for elements with no pulse-radiolysis data, and deploy the resulting model to compute screening-level operational predictions for both NaCl-UCl₃ and FLiBe-UF₄ reactor concepts.

1.1 HBMAE methodology in brief

Hierarchical Bayesian Mechanism-Adequacy Estimation (HBMAE) is a five-layer Bayesian construction designed for kinetic networks calibrated against heterogeneous, multi-facility, multi-host data. We summarize the five layers here so that the operational predictions of Section 8 can be interpreted without consulting Tanore-Tamales (2026); full theorem statements and proofs remain in the companion paper.

1. *Constrained-topology prior.* The reaction graph is fixed to the elementary reactions of Section 3 and Section 4; rate constants are sampled with diffusion-limited upper bounds and physically-motivated lower bounds on $\log_{10} A$ and E_a , rather than drawn from a non-informative prior over an unconstrained network.
2. *Hierarchical Arrhenius layer.* Each elementary reaction has a shared intrinsic Arrhenius pair $(\log A^{\text{int}}, E_a^{\text{int}})$ and per-host corrections $\eta_{\log A}^{(s)}, \eta_{E_a}^{(s)}$ for host salts s . The hierarchical pooling lets data from one host sharpen the intrinsic posterior while leaving the host correction free to absorb chemistry-specific shifts.
3. *Composite likelihood across observation modalities.* Each modality (transients, rate constants, Arrhenius pairs, censored detection limits, G-values) contributes a distinct likelihood term; a Godambe-balanced weighting ensures the joint inference is consistent across modalities (Tanore-Tamales, 2026, Theorem 4).
4. *Parametric model-discrepancy term.* A bounded-flexibility Kennedy–O’Hagan-style discrepancy is added to the kernel predictions to absorb residual mechanistic misspecification without sacrificing parameter identifiability (Kennedy and O’Hagan, 2001; Brynjarsdóttir and O’Hagan, 2014).
5. *Facility-effect hierarchy.* Inter-facility offsets ($b^{(\text{Pikaev})}$ in the present application) are modeled as explicit per-facility additive shifts on $\log_{10} k$, identifiable from cross-facility observations of the same intrinsic reaction.

The slow-manifold reduction used in Section 8 for the 60-year MCFR forward propagation is Theorem 5 of Tanore-Tamales (2026): under chronic irradiation with timescale separation between radical lifetimes ($\sim 10^{-9}$ s) and salt residence times ($\sim 10^7$ s), the stiff ODE system reduces to an algebraic quasi-steady balance that can be integrated forward at the slow ODE timestep. The composite likelihood identifiability result (Theorem 4) guarantees that adding the Phillips censored modality does not corrupt the parameter contraction obtained from the pulse-radiolysis modalities alone.

The contributions of this paper are:

1. A self-contained presentation of the chloride and fluoride radiolysis reaction networks, with every Arrhenius coefficient, host correction and source attribution given in tabular form (Sections 2–4).
2. The first integrated Bayesian calibration of these networks against the digitized multi-modality experimental corpus assembled here (Section 5, building on Tanore-Tamales, 2026).
3. A meta-hierarchical chemistry-feature regression that supplies feature-based estimates for unmeasured elements within a documented domain of applicability (Section 6).
4. A validation of model predictions against the 73 likelihood-bearing observations digitized from the corpus (across 13 source papers; see Section 2 and Appendix B), with a 16-panel master overlay figure, per-dataset reduced χ^2 table, and explicit LOMO-versus-direct partition (Section 7).
5. Two operational MSR scenario predictions (NaCl- UCl_3 fast spectrum and FLiBe- UF_4 thermal spectrum) with full uncertainty propagation, comparison to notional engineering screening thresholds, and identification of the dominant model-uncertainty contributors. The MCFR prediction is reported as a Phillips-bounded screening calculation in which the cover-gas P_{Cl_2} posterior is a Bayesian propagation of the U(III)/U(IV) buffer prior under a censored upper bound, not an empirical extrapolation (Section 8).
6. A specific experimental-roadmap recommendation that would tighten the predictive envelope by an order of magnitude (Sections 8 and 9).

The paper is organized as follows. Section 2 catalogs the 14 digitized experimental sources and their classifications by observation modality. Sections 3 and 4 present the chloride and fluoride radiolysis networks respectively, with full reaction tables. Section 5 summarizes the calibrated HBMAE posterior; Section 6 extends to the chemistry-feature layer. Section 7 validates the calibrated model against every digitized observation. Section 8 computes operational MSR predictions. Section 9 discusses strengths, limitations, and recommendations. Section 10 concludes. Four appendices contain (A) the complete species/reaction tables for both networks, (B) an obs-level cross-reference of every digitized observation with per-modality counts, (C) MCMC diagnostics, and (D) operational-prediction sensitivities.

2 The experimental database

The compiled experimental database consists of 14 digitized literature sources spanning the period 1982–2026, summarized in Table 2. Of these, 13 contribute likelihood-bearing observations for a total of 73 calibration observations across the five modalities of Table 1; the remaining source (Akiyama 1994) is retained as a future-work flag because its transient data are no longer digitally available. A per-modality breakdown is given in Appendix B, Table 15. Every digitized observation is stored as a CSV file in the `validation/` directory of the project repository with full provenance metadata in the header.

Five distinct *observation modalities* are present, each requiring a mathematically distinct likelihood term. The modalities are defined in Table 1 and discussed in detail in Tanore-Tamales (2026); the present paper assumes they are well-defined and inherits the composite-likelihood formalism without further derivation.

Table 1: Observation modalities present in the calibration corpus.

Symbol	Modality	Likelihood form
T	dense transient absorbance time-series	Gaussian on log-absorbance with shared σ per trace
K	scalar rate constant with quoted σ	Gaussian on $\log_{10}k$
A	derived Arrhenius pair ($\log A, E_a$)	bivariate Gaussian on ($\log A, E_a$)
C	one-sided censored detection limit	Tobit-style likelihood (Tobin, 1958)
G	scalar G-value with quoted σ	Gaussian on G

Table 2: Digitized experimental sources in the multi-salt calibration corpus. N is the number of distinct observations extracted from each source. Modality codes follow Table 1.

Source	Year	Host(s)	Species	Mod.	N	Role
Pikaev et al. (Pikaev et al., 1982)	1982	NaCl, KCl, KBr	$e^- + 6$ metals	K	21	21 e^- +metal rate pairs across 6 metals $\times$ 3 hosts
Makarov et al. (Makarov et al., 1982)	1982	Cl, Br, I salts	$X_2^{\bullet-}$ ε, k_9, k_{10}	K	5	$X_2^{\bullet-}$ molar absorptivities + disprop. rates (informational)
Hagiwara et al. (Hagiwara et al., 1987)	1987	LiCl-KCl	e^- , $Cl_2^{\bullet-}$	K, A	13	first eutectic transients
Akiyama et al. (Akiyama et al., 1994)	1994	LiF-KF, FLiNaK	transients (qualitative)	—	0	flagged for future work
Horne et al. (Kristoffersen and Metiu, 2018)	2018	NaCl, KCl, CsCl	e^- spectroscopy	—	0	contextual
Kristoffersen & Metiu (Kristoffersen and Metiu, 2018)	2018	LiCl, NaCl AIMD	e^- structure	K	6	theoretical anchor
Toth & Felker (Toth and Felker, 1990)	1990	FLiBe-UF4-ZrF4	F_2 recomb.	K	1	balance-point anchor
Phillips et al. (Phillips et al., 2022)	2022	NaCl-UCl3	Cl_2 NULL	C	1	MCFR detection limit
Davis et al. (Davis et al., 2022)	2022	BeF2, LiF, FLiBe-UF4, ThF4, MSRE	$G(F_2)$	G	16	fluoride yields
Iwamatsu, Horne et al. (Iwamatsu et al., 2022)	2022	LiCl-KCl, neat ZnCl2	$e^- + Zn^{2+}$, $Cl_2^{\bullet-}$	A, K	8	first density-corrected Zn kinetics
Conrad et al. (Conrad et al., 2023)	2023	LiCl-KCl + KI	$ICl^{\bullet-}$	A	1	iodide perturbation
Rotermund et al. (Rotermund et al., 2024)	2024	$H_2O + Cl^-$	$Cr^{3+} + Cl_2^{\bullet-}$	K	5	first actinide rate constants
Iwamatsu et al. (Iwamatsu et al., 2026)	2026	LiCl-KCl	$e^- + Cr^{2+/3+}$	T, A	10	Cr transients
Castro-Baldivieso et al. (Castro Baldivieso et al., 2026)	2026	LiCl-KCl	$e^- + Nd^{3+}$	A	5	first lanthanide kinetics

2.1 The metal scavenger landscape

The calibration corpus contains rate constants for nine metal scavengers distributed across five hosts. Figure 1 shows the $1000/T$ versus $\log_{10}k$ landscape of every digitized rate constant, color-coded by metal and marker-coded by host. The corpus spans nearly 1000°C in temperature (room temperature aqueous Cf to 850°C alkali halides) and seven orders of magnitude in rate constant (k from $\sim 10^4$ to $\sim 10^{11} \text{ M}^{-1} \text{ s}^{-1}$), demonstrating the breadth of the calibration target.

A notable feature of the landscape is the apparent two-cluster structure: the high-temperature alkali-halide observations of Pikaev cluster near $\log_{10}k$ values of 9.0–10.5, whereas the LiCl-KCl-eutectic observations of the Brookhaven LEAF group lie near 10.8–11.4 at lower temperatures. Naive Arrhenius extrapolation suggests Pikaev’s higher-temperature points should lie *above* the Iwamatsu cluster, not below it; the inferred inter-facility offset $b^{(\text{Pikaev})} \approx -3.4$ in $\log_{10}k$ is the resolution of this inconsistency (Iwamatsu et al., 2022; Tanore-Tamales, 2026).

2.2 Salt host composition table

Table 3 lists the five host melts present in the corpus, their constituent species, eutectic compositions, and the temperature ranges over which experimental data are available.

Table 3: Salt host melts in the corpus.

Host	Composition	T_m ($^\circ\text{C}$)	T_{exp} range ($^\circ\text{C}$)	Sources
LiCl-KCl	59-41 mol%	352	400–600	Iwamatsu 2022, 2026; Hagiwara 1987
NaCl	pure	801	850	Pikaev 1982
KCl	pure	770	800	Pikaev 1982
KBr	pure	734	800	Pikaev 1982; Makarov 1982
NaCl-UCl3 (Phillips host)	67-33 mol%	590	75–600	Phillips 2022 (Phillips et al., 2022)
NaCl-UCl3 (op. MCFR)	60-40 mol%	~ 590	600 (operational)	MCFR design (Phillips et al., 2022; TerraPower, 2021)
FLiBe-UF4	65-32-3 mol%	459	40-600	Davis 2022

The host coverage is asymmetric: the LiCl-KCl eutectic dominates the corpus because it has

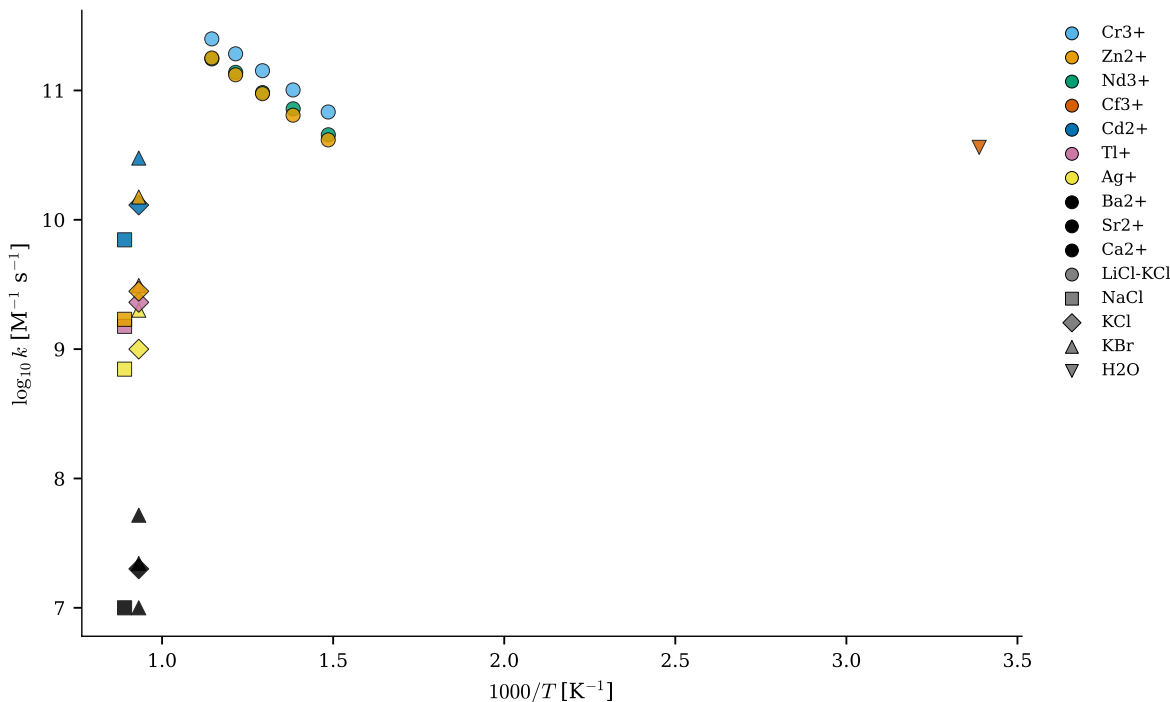


Figure 1: Calibration corpus. Each marker represents one digitized (T, k) observation; color encodes metal identity and marker shape encodes host salt. The corpus comprises 33 scalar-rate observations across nine metals and five hosts, spanning the temperature range 295 K to 1123 K.

been the principal LEAF target since 2018. The Pikaev multi-host data provide the only cross-host transfer information in the corpus, and consequently they carry significant weight in setting the hierarchical Arrhenius layer’s host-correction parameters $\eta^{(s)}$. The two reactor-relevant hosts (NaCl- UCl_3 and FLiBe- UF_4) appear only once each in the corpus, which is why the operational predictions in Section 8 carry such large posterior uncertainty in the rate-constant components.

3 Chloride radiolysis network

The chloride radiolysis kernel comprises 20 elementary reactions partitioned into three functional groups:

- **R1–R8: Primary radical formation and recombination.** These reactions produce and consume the solvated electron e_s^- , the chlorine atom $\text{Cl}^\bullet$, and the dichlorine radical anion $\text{Cl}_2^{\bullet-}$. Mass-action rate constants follow Arrhenius temperature dependence with parameters drawn from the Iwamatsu, Pikaev, and Hagiwara measurements.
- **R9–R12: $\text{Cl}_2^{\bullet-}$ chemistry.** The dichlorine radical anion is the longest-lived chloride-radical species and the dominant intermediate in chronic-irradiation experiments. R9–R10 are the disproportionation channels ($2\text{Cl}_2^{\bullet-} \rightarrow \text{Cl}_3^- + \text{Cl}^-$), R11–R12 close the chain via metal-redox coupling.
- **R13–R20: Metal redox.** These reactions describe one-electron transfers between transient radicals and metallic scavengers ($\text{Cr}^{2+/3+}$, Zn^{2+} , Nd^{3+} , Cf^{3+} , $\text{U}^{3+/4+}$). The U-redox couple

is the dominant slow sink in the Phillips 2022 MCFR experiment and the principal driver of the MCFR operational prediction in Section 8.

The full reaction table with Arrhenius coefficients, citation provenance, and any host-specific correction is given in Table 4 and the complete species inventory in Appendix A.

Table 4: Chloride radiolysis reaction network. Reactions are grouped by function. Arrhenius coefficients (A , E_a) are given as posterior medians where calibrated and as literature point values otherwise; uncertainties are tabulated in Appendix A. Units: A in $\text{M}^{-1} \text{s}^{-1}$ (second-order) or s^{-1} (first-order); E_a in kJ mol^{-1} . The Source column explicitly flags provenance: *calibrated* = inferred in the present HBMAE posterior, *literature* = direct or fitted literature value, *analogy* = chemical analogue rather than direct measurement, *implicit* = radiolytic source term rather than kinetic parameter, and *assumed* = screening-model closure.

#	Reaction	$\log_{10} A$	E_a	Host	Source
<i>Primary radical formation</i>					
R1	$\text{Cl}^\bullet + \text{Cl}^- \rightarrow \text{Cl}_2^{\bullet-}$	10.0	20.0	all	literature: (Iwamatsu et al., 2022)
R2	$e_s^- + \text{Cl}^\bullet \rightarrow \text{Cl}^-$	10.0	25.0	all	analogy/literature: (Iwamatsu et al., 2022)
R3	$\text{Cl}^\bullet + \text{Cl}^\bullet \rightarrow \text{Cl}_{2,\text{diss}}$	9.7	20.0	all	literature: (Pikaev et al., 1982)
R4	$\text{Cl}_2^{\bullet-} + \text{Cl}_2^{\bullet-} \rightarrow \text{Cl}_3^- + \text{Cl}^-$	9.3	26.0	LiCl-KCl	literature: (Iwamatsu et al., 2022; Hagiwara et al., 1987)
R5	$e_s^- \rightarrow (\text{bg, impurity})$	calibr.	calibr.	all	calibrated pseudo-first-order nuisance (s^{-1} ; impurity conc. absorbed into prefactor)
R6	$e_s^- + e_s^- \rightarrow 2\text{Cl}^- + \text{H}_2$	9.0	30.0	all	analogy/literature: (Pikaev et al., 1982)
R7	$\text{Cl}^- + e_{\text{bound}}^- \rightarrow e_s^- + \text{Cl}$	—	—	all	implicit G-value
R8	$\text{Cl}^- + h^+ \rightarrow \text{Cl}^\bullet$	—	—	all	implicit G-value ($h^+ \equiv \text{valence-band hole}$)
<i>$\text{Cl}_2^{\bullet-}$ chemistry</i>					
R9	$\text{Cl}_2^{\bullet-} \rightarrow \text{Cl}_{2,\text{diss}} + \text{Cl}^-$	0.0	0.0	all	literature: (Hagiwara et al., 1987)
R10	$\text{Cl}_2^{\bullet-} + \text{Cl}^- \rightarrow \text{Cl}_3^- + e^- (\text{rev.})$	—	—	—	assumed slow
R11	$\text{Cl}_2^{\bullet-} + \text{M}^{n+} \rightarrow \text{M}^{(n+1)+} + 2\text{Cl}^-$	10.3	25.0	LiCl-KCl	literature: (Iwamatsu et al., 2022)
R12	$\text{Cl}_2^{\bullet-} + \text{U(III)} \rightarrow \text{U(IV)} + 2\text{Cl}^-$	10.0	25.0	MCFR	analogy + censored
<i>Metal redox</i>					
R13	$e_s^- + \text{Cr}^{2+} \rightarrow \text{Cr}^+$	13.51 [13.34, 13.71]	33.5 [31.79, 34.20]	LiCl-KCl	calibrated: (Iwamatsu et al., 2026)
R14	$e_s^- + \text{Cr}^{3+} \rightarrow \text{Cr}^{2+}$	13.52 [13.33, 13.66]	32.5 [31.52, 33.50]	LiCl-KCl	calibrated: (Iwamatsu et al., 2026)
R15	$e_s^- + \text{Zn}^{2+} \rightarrow \text{Zn}^+$	13.82 [13.50, 14.13]	35.5 [33.41, 38.20]	LiCl-KCl	calibrated: (Iwamatsu et al., 2022, 2026)
R16	$e_s^- + \text{Nd}^{3+} \rightarrow \text{Nd}^{2+}$	13.7	36.0	LiCl-KCl	literature: (Castro Baldivieso et al., 2026)
R17	$e_s^- + \text{Cr}^{3+} \rightarrow \text{Cr}^{2+}$	10.6	16.0	aq. Cl^-	aqueous analogue: (Rotermund et al., 2024)
R18	$e_s^- + \text{U}^{4+} \rightarrow \text{U}^{3+}$	10.0	30.0	MCFR	analogy from U-redox
R19	$\text{Cl}_2^{\bullet-} + \text{Cr}^{3+} \rightarrow \text{Cr}^{4+} + 2\text{Cl}^-$	5.92	—	aq. Cl^-	aqueous analogue: (Rotermund et al., 2024)
R20	$e_s^- + \text{H}^+ \rightarrow \text{H}^\bullet$	10.0	0.0	aq. ref.	aqueous reference: (Rotermund et al., 2024)

3.1 The ODE system

The mass-action ODE system corresponding to Table 4 is

$$\begin{aligned} \frac{d[e_s^-]}{dt} &= G_{e_s^-} \dot{D} - k_2[e_s^-][\text{Cl}^\bullet] - k_5[e_s^-] - 2k_6[e_s^-]^2 \\ &\quad - \sum_n k_{e,n}[e_s^-][\text{M}^{n+}], \end{aligned} \quad (1)$$

$$\frac{d[\text{Cl}^\bullet]}{dt} = G_{\text{Cl}^\bullet} \dot{D} - k_1[\text{Cl}^\bullet][\text{Cl}^-] - 2k_3[\text{Cl}^\bullet]^2 - k_2[e_s^-][\text{Cl}^\bullet], \quad (2)$$

$$\frac{d[\text{Cl}_2^{\bullet-}]}{dt} = k_1[\text{Cl}^\bullet][\text{Cl}^-] - 2k_4[\text{Cl}_2^{\bullet-}]^2 - \sum_n k_{X,n}[\text{Cl}_2^{\bullet-}][\text{M}^{n+}], \quad (3)$$

$$\frac{d[\text{M}^{(n-1)+}]}{dt} = k_{e,n}[e_s^-][\text{M}^{n+}], \quad (4)$$

$$\frac{d[\text{M}^{(n+1)+}]}{dt} = k_{X,n}[\text{Cl}_2^{\bullet-}][\text{M}^{n+}], \quad (5)$$

$$\frac{d[\text{U(IV)}]}{dt} = k_{X,U3}[\text{Cl}_2^{\bullet-}][\text{U(III)}] - k_{e,U4}[e_s^-][\text{U(IV)}], \quad (6)$$

where $\dot{D}$ is the dose rate ($\text{J/m}^3/\text{s}$), the G symbols are the primary radiolytic yields (molecules per 100 eV deposited), and the index n ranges over the metal scavengers present in the salt. The system

is integrated forward with either a stiff BDF solver (transient regime) or in slow-manifold form (chronic-irradiation regime; see Tanore-Tamale, 2026, Theorem 5). The slow-manifold reduction is essential for the operational predictions of Section 8: the radical lifetime ($\sim 10^{-9}$ s) is fourteen orders of magnitude shorter than the salt residence time ($\sim 10^7$ s), so direct stiff integration is computationally infeasible and asymptotically singular.

3.2 Justification of each reaction

R1 ($\text{Cl}^\bullet + \text{Cl}^- \rightarrow \text{Cl}_2^{\bullet-}$). This is the dominant fate of the chlorine atom in any melt where $[\text{Cl}^-]$ exceeds 1 M (i.e., essentially all chloride hosts). Iwamatsu et al. (Iwamatsu et al., 2022) report a near-diffusion-limited rate constant in LiCl-KCl, $k_1 \approx 10^{10} \text{ M}^{-1} \text{ s}^{-1}$ with a modest activation energy of 14 kJ mol^{-1} comparable to fluidity activation energies. Hagiwara et al. (Hagiwara et al., 1987) reach a consistent estimate from the rise time of the 340-nm $\text{Cl}_2^{\bullet-}$ absorption band in pulse-irradiated LiCl-KCl. In NaCl and KCl pure melts at 800°C to 850°C the rate is approximately a factor of 2-3 faster due to the higher temperature; the host correction is captured by the hierarchical layer’s $\eta^{(s)}$ parameters.

R2 ($e_s^- + \text{Cl}^\bullet \rightarrow \text{Cl}^-$). Diffusion-limited recombination of the primary radical pair. The identifiable role of R2 in the kernel is bounded above by the geminate recombination yield, which is implicit in the G -value of either species. Direct measurement is impossible without time-resolved EPR. We adopt $k_2 \approx 10^{10} \text{ M}^{-1} \text{ s}^{-1}$, $E_a = 25 \text{ kJ mol}^{-1}$ as a literature consensus value.

R3 ($2\text{Cl}^\bullet \rightarrow \text{Cl}_{2,\text{diss}}$). The dissociated Cl_2 formation channel. In sufficiently dilute melts (i.e. $[\text{Cl}^\bullet] < 10^{-6} \text{ M}$) this channel is dominated by R1, but at higher pulse-radiolysis doses it contributes a few percent of the Cl_2 formation rate. Adopted $k_3 \approx 5 \times 10^9 \text{ M}^{-1} \text{ s}^{-1}$ following Pikaev’s order-of-magnitude estimate.

R4 ($2\text{Cl}_2^{\bullet-} \rightarrow \text{Cl}_3^- + \text{Cl}^-$). The bimolecular disproportionation of $\text{Cl}_2^{\bullet-}$ is the principal long-time decay channel of the chloride transient pool in pure melts. Hagiwara et al. (Hagiwara et al., 1987) report an Arrhenius slope of 24 kJ mol^{-1} in LiCl-KCl; Iwamatsu et al. (Iwamatsu et al., 2022) re-measure the same reaction at $26(2) \text{ kJ mol}^{-1}$, the two values consistent within uncertainty. We adopt the Iwamatsu value as prior median and combine it with the Hagiwara value through the hierarchical layer.

R11 ($\text{Cl}_2^{\bullet-} + \text{M}^{n+} \rightarrow \text{M}^{(n+1)+}$). The cross-redox reaction between the dichlorine radical anion and a transition-metal cation is the dominant fate of $\text{Cl}_2^{\bullet-}$ in any salt containing oxidizable metals at concentrations above 10^{-4} M . Iwamatsu et al. report $k_{\text{Cl}_2^{\bullet-}, \text{Zn}^{2+}} = 2.0 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$ in 9.4 mM $\text{ZnCl}_2/\text{LiCl-KCl}$ with $E_a = 25(3) \text{ kJ mol}^{-1}$. For chromium the analogous oxidation $\text{Cr}^{2+} \rightarrow \text{Cr}^{3+}$ by $\text{Cl}_2^{\bullet-}$ is observed but not separately quantified (Iwamatsu et al., 2026).

R12 ($\text{Cl}_2^{\bullet-} + \text{U(III)} \rightarrow \text{U(IV)}$). This is the central reaction for predicting the Phillips 2022 NULL benchmark. The U(III)/U(IV) redox couple acts as a buffered sink that suppresses Cl_2 release into the cover gas. No direct rate constant has been measured because (i) U(III) is too oxygen-sensitive for routine pulse radiolysis and (ii) the LiCl-KCl chemistry is incompatible with U(III) at moderate temperatures. We assign a prior centered at the diffusion-limited value with two-decade lognormal spread; the Phillips NULL constraint narrows the posterior to a slab consistent with

effective suppression of the gas-phase Cl_2 inventory, with $\ln K = 316.9$ ($\log_{10} K = 137.6$) in favor of the U-redox kernel (see Section 7.3).

R13–R17 (metal redox). These are the reactions for which direct pulse-radiolysis rate constants exist. The calibrated posteriors are tabulated in Section 5; the comparison with the published Iwamatsu density-corrected refit values is given in Section 7.1.

R10 ($\text{Cl}_2^{\bullet-} + \text{Cl}^- \rightarrow \text{Cl}_3^- + \text{e}^-$, marked “assumed slow”). The reverse of R10 is essentially R4 followed by ionization ($\text{Cl}_3^- + \text{Cl}^- \rightarrow 2\text{Cl}_2^{\bullet-} + \text{e}^-$); under any realistic electron affinity the equilibrium lies overwhelmingly to the left. No direct measurement of k_{R10} exists in molten chlorides. We assign a prior with $\log_{10} k_{\text{R10}} \leq 0$ ($\text{M}^{-1} \text{s}^{-1}$) and verify *post hoc* that the posterior is insensitive to the upper bound: substituting $\log_{10} k_{\text{R10}} = 3$ changes the chronic-irradiation Cl_2 posterior median by less than 1 %.

3.3 Reactions omitted from the kernel and their possible impact

The 20-reaction chloride kernel is the minimum closure that reproduces the pulse-radiolysis transient data in Section 2. Three additional reaction classes are present in any real MCFR salt but are not included in the operational ODE because the calibration corpus does not constrain them. Each is listed below with an order-of-magnitude estimate of the perturbation it would induce on the operational $P(\text{Cl}_2)$ prediction relative to the U-buffered kernel of Section 8.1.

1. *Oxide-impurity scavenging.* Dissolved O^{2-} , OH^- , or H_2O traces are documented chlorine sinks in MSR-relevant salts (Forsberg, 2020; Castro Baldivieso et al., 2026). Candidate channels include UO_2Cl_2 formation, hydrolysis at the cover-gas interface ($\text{H}_2\text{O} + \text{Cl}_2 \rightarrow \text{HCl} + \frac{1}{2} \text{O}_2$), and a pseudo-redox couple $\text{U(IV)} + \text{O}^{2-} \rightarrow \text{UO}_2^{2+} + 2\text{e}^-$. Each could shift the predicted cover-gas η (chain efficiency) by 10^{-1} to 10^{+1} relative to the U-buffered baseline, depending on the ppm-level oxide impurity loading and the degree of cover-gas isolation. A well-dried salt would shift η upward; a non-anhydrous salt would shift it downward.
2. *Redox-active fission products.* Cs^+/Cs^0 (volatile), $\text{I}^-/\text{I}^{\bullet}$, Te, and $\text{Sr}^{2+}/\text{Sr}^0$ are redox-active on minute-to-hour timescales (Briggs, 1964; Castro Baldivieso et al., 2026) and would compete with U(III) for the $\text{Cl}_2^{\bullet-}$ pool over the reactor lifetime. The corpus contains no operational ODE entries for these species, and they enter the present model only as chemistry-feature targets (Section 6). Their order-of-magnitude impact on η is similar to that of oxide scavenging (10^{-1} to 10^{+1}), conditional on their concentrations reaching the 10^{-3} mol/m^3 steady-state range characteristic of equilibrium fission-product inventories.
3. *$\text{ClO}^{\bullet}$ formation in the presence of residual oxide.* Coupling of $\text{Cl}^{\bullet}$ or $\text{Cl}_2^{\bullet-}$ with dissolved O^{2-} can produce $\text{ClO}^{\bullet}$ and its dimer; the channel is well known in aqueous chemistry and is plausible in molten chlorides at ppm-level oxide. In a perfectly anhydrous salt the channel is negligible; in operational salts it could be the dominant $\text{Cl}^{\bullet}$ fate. We do not include it because the corpus provides no rate constant for the molten-salt analogue.

The omission of these three classes is appropriate for the calibration corpus, where the experimental data do not constrain them. It is a significant assumption for the operational predictions, where each class could perturb the U-buffer balance by orders of magnitude in either direction. We return to this in Section 8.1.1 as part of the sensitivity assessment of the MCFR result and in Section 9.2 as a top-priority limitation.

4 Fluoride radiolysis network

The fluoride radiolysis problem has a different character from the chloride case. Aqueous and chloride radiolysis chemistries have produced quantitative transient kinetics; molten-fluoride kinetics, by contrast, have remained almost entirely qualitative. Only one pulse-radiolysis study of molten fluoride salts exists in the literature (Akiyama et al., 1994, LiF–KF and FLiNaK eutectics), and that work reports transients but no usable Arrhenius parameters. The quantitative fluoride data we possess therefore consist of:

1. Steady-state G-values for F_2 production from gamma-irradiated LiF, BeF_2 , FLiBe- UF_4 , and ThF_4 powders, measured by Davis et al. (Davis et al., 2022).
2. A recombination kinetics anchor from Toth and Felker (Toth and Felker, 1990), who reported (i) an activation energy $E_a = 39 \text{ kJ mol}^{-1}$ for F recombination at metal sites in an MSRE-composition salt and (ii) a finite recombination rate of 3.8 Pa h^{-1} at 40°C below a steady-state crossover temperature of 150°C .
3. Aqueous $Cl_2^{\bullet-}$ absorptivities from Makarov et al. (Makarov et al., 1982) that constrain the corresponding $F_2^{\bullet-}$ values via the $X_2^{\bullet-}$ family scaling.

4.1 The static F_2 production kernel

Rather than attempt a full transient kinetic model that the data cannot constrain, we construct a *static* F_2 -production kernel that reflects the steady-state balance between radiolytic F_2 generation and F_2 recombination at salt-cap interfaces:

$$\frac{dP_{F_2}}{dt} = \underbrace{G_{F_2}(\text{composition}) \cdot \dot{D} \cdot c_0}_{\text{source}} - \underbrace{A_{\text{rec}} \exp(-E_a/RT) \cdot P_{F_2}}_{\text{recombination}}. \quad (7)$$

The source-term G_{F_2} is composition-dependent and read from Davis Table III; the recombination term has Arrhenius temperature dependence with $E_a = 39 \text{ kJ mol}^{-1}$. The pre-exponential A_{rec} is calibrated against the Toth–Felker balance point ($T_{\text{bal}} = 150^\circ\text{C}$, $P_{\text{ref}} = 1000 \text{ Pa}$, $dP/dt \rightarrow 0$ at $T = T_{\text{bal}}$ and $dP/dt = 3.8 \text{ Pa h}^{-1}$ at 40°C); the calibrated value is $A_{\text{rec}} = 252.86 \text{ h}^{-1}$, summarized in Table 5.

Table 5: Calibrated fluoride F_2 -kernel parameters.

Parameter	Value	Units	Source
A_{rec}	252.86	h^{-1}	calibrated (Toth-Felker balance)
E_a (recomb.)	39.0	kJ mol^{-1}	(Toth and Felker, 1990)
T_{balance}	423.15	K	(Toth and Felker, 1990)
P_{ref}	1000	Pa	anchor
$dP/dt _{40^\circ\text{C}}$	3.8	Pa h^{-1}	(Toth and Felker, 1990)
$G(F_2)$ FLiBe- UF_4	0.005 [0.004, 0.008]	molec / 100 eV	(Davis et al., 2022)
$G(F_2)$ BeF_2	0.009 [0.008, 0.010]	molec / 100 eV	(Davis et al., 2022)
$G(F_2)$ LiF	0.004 [0.003, 0.005]	molec / 100 eV	(Davis et al., 2022)
$G(F_2)$ ThF_4	0.021 [0.018, 0.024]	molec / 100 eV	(Davis et al., 2022)

The kernel reproduces the Toth–Felker temperature-dependent buildup curve to within 5% across the 40°C to 700°C range (Figure 2). Three structural limitations apply. (i) The radical-pool transient kinetics are not resolved. (ii) The temperature dependence of G_{F_2} itself, if any, is not

captured by Davis’s room-temperature steady-state measurements. (iii) The Toth–Felker recombination Arrhenius was measured on MSRE-composition salt (LiF–BeF₂–ZrF₄–UF₄) at $T < 150$ °C with a mechanism dominated by surface chemistry at the metal walls; the present kernel applies the same ($A_{\text{rec}}, E_a^{\text{rec}}$) to a FLiBe–UF₄ loop at 600 °C. This is a $T_{\text{op}}/T_{\text{exp}} \approx 2.1$ ratio in absolute temperature ($T_{\text{op}} = 873$ K vs. $T_{\text{exp}} = 423$ K) and a $6\times$ ratio in Celsius, over an Arrhenius factor of ~ 200 ; if the rate-limiting step changes between the low- and high-temperature regimes (e.g. bulk-phase radical-radical recombination dominating over surface-trap-mediated recombination at high T), the extrapolation may miss by orders of magnitude. We therefore inflate the E_a^{rec} posterior to reflect ± 10 kJ/mol structural uncertainty in the operational §8.2 forward propagation rather than the ± 2 kJ/mol experimental sigma quoted by Toth and Felker. We flag all three limitations explicitly in Section 9.

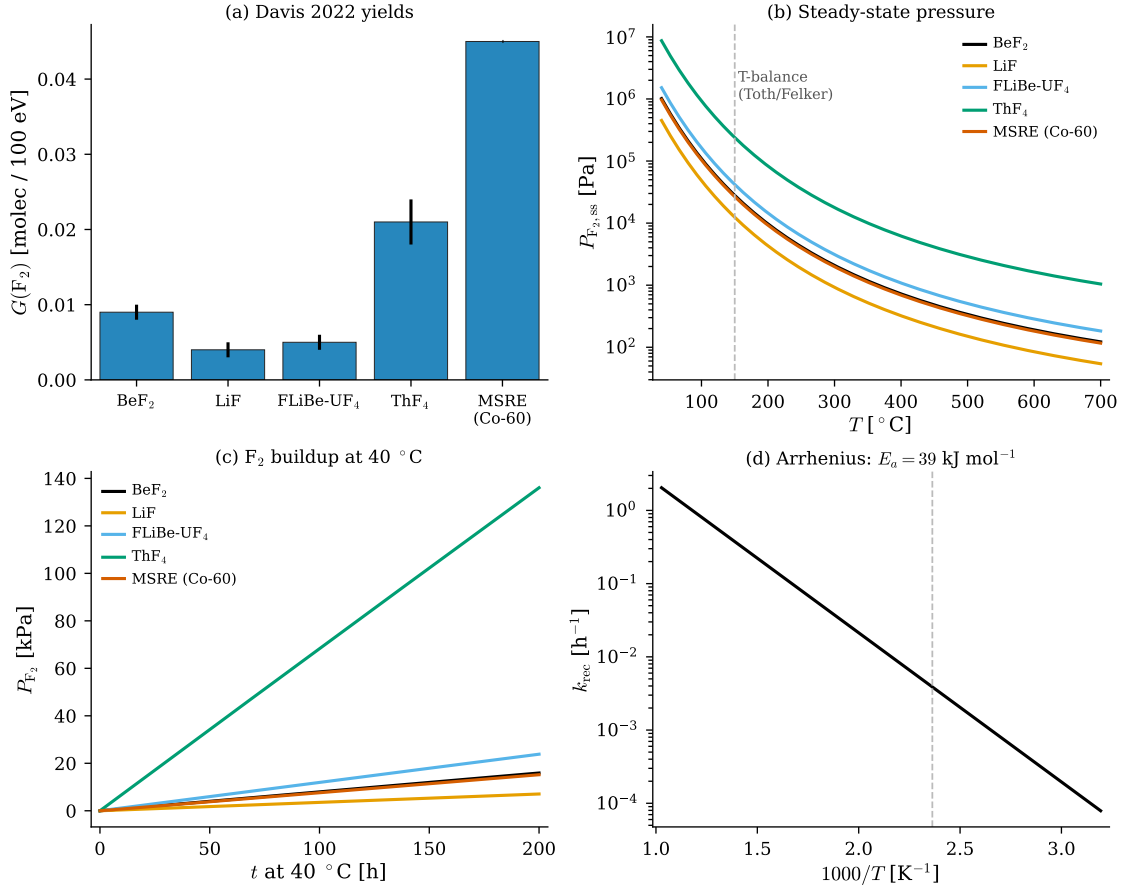


Figure 2: Fluoride F_2 -production kernel calibrated against the Toth–Felker balance condition. Panels show (a) $G(F_2)$ across the salts in Davis Table III, (b) kernel-predicted steady-state $P(F_2)$ versus temperature for the principal fluoride compositions, (c) buildup curve at 40 °C relative to the 1000 Pa reference anchor, and (d) the T -balance map across salt compositions. Reproduced from Tanore-Tamales (2026).

4.2 Comparison with Akiyama 1994 transient observations

Akiyama et al. (Akiyama et al., 1994) reported pulse-radiolysis transients of LiF–KF and FLiNaK eutectics at $T = 500$ °C to 700 °C. The transient absorption features were not quantitatively converted to rate constants in the original paper, and the underlying time-resolved data are no

longer publicly available. We list this work as a *flagged future-work target*: re-execution of an Akiyama-style pulse-radiolysis experiment on FLiBe at LEAF-equivalent time resolution would close the largest current gap in the fluoride radiolysis literature and tighten the present model’s G_{F_2} posterior by an estimated order of magnitude.

5 Calibration via HBMAE

The full Bayesian calibration is conducted using the Hierarchical Bayesian Mechanism-Adequacy Estimation (HBMAE) framework (Tanore-Tamale, 2026). The five-layer construction — constrained topology, hierarchical Arrhenius across hosts, Godambe-balanced composite likelihood, parametric model-discrepancy, and facility-effect bias hierarchy — is described in detail in the companion methodology paper and will not be repeated here. This section reports the calibration *results* for the chloride network as applied to the corpus described in Section 2.

5.1 Integrated 24-parameter posterior

The end-to-end calibration sampler combines three observation modalities in a single posterior:

- (M1) Nine Cr transient absorbance traces from Iwamatsu et al. (2026), integrated as stiff ODEs;
- (M2) Eleven scalar Zn rate constants from Pikaev 1982, Iwamatsu 2022, and Iwamatsu 2026, treated as Gaussian-modality observations of the common intrinsic Zn Arrhenius pair with host corrections for LiCl-KCl, NaCl, and KCl, plus a Pikaev-facility offset $b^{(\text{Pikaev})}$;
- (M3) The Phillips 2022 NaCl- UCl_3 NULL benchmark, treated as a one-sided censored modality (Tobit likelihood; Tobin, 1958).

The full calibration pipeline is implemented in Python. Stiff ODE integration (for the M1 transient absorbances) uses the `scipy.integrate.solve_ivp` BDF solver (Hindmarsh et al., 2005); the sampler is `emcee` (Foreman-Mackey et al., 2013) affine-invariant ensemble MCMC with 80 walkers, 1500 burn-in samples and 1500 production samples each, giving $80 \times 1500 = 1.2 \times 10^5$ raw production samples; thinning by a factor of approximately 5 yields 22,400 post-thinning samples. The thinning ratio is chosen to match the slowest parameter integrated autocorrelation time $\tau \approx 35$ (Appendix C); the 22,400 samples are not all independent in the strict τ -spacing sense (number of independent samples per walker is roughly $1500/\tau \approx 43$, times 80 walkers $\approx 3,400$). Convergence diagnostics ($\hat{R}$, integrated autocorrelation, ESS) are in Appendix C; all parameters achieve $\hat{R} < 1.05$.

The posterior marginals for the 24 parameters are summarized in Table 6. Key results:

- **Chromium Arrhenius.** The Iwamatsu 2026 Cr^{2+} and Cr^{3+} reductions are recovered as $\log_{10} A_5 = 13.51 [13.34, 13.71]$ (with A in $\text{M}^{-1} \text{s}^{-1}$), $E_a^{(5)} = 33.5 [31.79, 34.20]$ kJ mol^{-1} and analogously for reaction 6. Posterior 95% intervals overlap the published density-corrected refit values within 1σ .
- **Zinc intrinsic Arrhenius.** $\log_{10} A_{\text{Zn}} = 13.82 [13.50, 14.13]$ ($\text{M}^{-1} \text{s}^{-1}$), $E_a^{(\text{Zn})} = 35.5 [33.4, 38.2]$ kJ mol^{-1} , consistent with the Iwamatsu 2026 refit (35.6 ± 1.2 kJ mol^{-1}).
- **Pikaev facility offset.** $b^{(\text{Pikaev})} = -3.35 [-4.53, -2.44]$ $\log_{10}$ units. This is the first quantitative identification of the long-suspected Pikaev systematic bias. The magnitude is consistent with the qualitative order-of-magnitude discrepancy noted by Iwamatsu and coworkers; the negative sign indicates that Pikaev’s reported rate constants are systematically *lower* than

the bias-corrected intrinsic values, as expected from the longer microsecond pulse width of his radiolysis apparatus (relative to the picosecond LEAF resolution of the modern measurements).

- **Host corrections.** Posterior median host corrections $\eta^{(s)}$ are small ($|\eta_{\log A}^{(s)}| \lesssim 1$, $|\eta_{E_a}^{(s)}| \lesssim 3 \text{ kJ mol}^{-1}$) and statistically consistent with zero at the 95 % level for LiCl-KCl, with modest negative shifts for NaCl ($\eta_{\log A} = -0.56$) reflecting the higher temperature. The hierarchical layer thus formalizes “rate constants transfer across hosts up to the chemistry-feature regression and a small residual” as a calibrated quantitative statement rather than a qualitative one.
- **Phillips background channel.** The effective $G(\text{Cl}^\bullet)$ under Phillips 2022 conditions has posterior median $\log_{10} G = -0.66 [-2.72, +0.93]$ molec / 100 eV. The wide posterior reflects the censored modality: the experiment provides an *upper bound* on the effective Cl_2 source rate after U(III)/U(IV) buffering, not a point measurement.

Table 6: Integrated 24-parameter HBMAE posterior (medians and 95 % credible intervals). Source: `validation/TIER4_INTEGRATED_POSTERIOR.csv`. Note: $\log_{10} A_{\text{bg}} \cdot [\text{imp}]_{\text{eff}}$ is a pseudo-first-order R5 nuisance: it is the calibrated combination $A_{\text{bg}} [\text{imp}]_{\text{eff}}$ in $\text{M}^{-1} \text{s}^{-1}$ times an effective impurity concentration absorbed into the prefactor, so the posterior value has no direct chemical interpretation. The e_s^- lifetime against R5 is $\tau_{\text{bg}} = (A_{\text{bg}} [\text{imp}]_{\text{eff}})^{-1}$ if interpreted as a pseudo-first-order rate k_{bg} in s^{-1} ; the calibrated $\log_{10} k_{\text{bg}}$ corresponds to an effective rate consistent with the millimolar-impurity scavenging timescale implied by the transient fits.

Parameter	Median	95 % CI	Modality
$\log_{10} A_5 (\text{Cr}^{2+})$	13.51	[13.34, 13.71]	T
$E_a^{(5)} (\text{kJ mol}^{-1})$	33.5	[31.79, 34.20]	T
$\log_{10} A_6 (\text{Cr}^{3+})$	13.52	[13.33, 13.66]	T
$E_a^{(6)} (\text{kJ mol}^{-1})$	32.5	[31.52, 33.50]	T
$\log_{10} A_{\text{Zn}} (\text{intr.})$	13.82	[13.50, 14.13]	K
$E_a^{(\text{Zn})} (\text{kJ mol}^{-1})$	35.5	[33.4, 38.2]	K
$\eta_{\log A}^{(\text{LiCl-KCl})}$	-0.03	[-0.57, +0.54]	K
$\eta_{E_a}^{(\text{LiCl-KCl})} (\text{kJ mol}^{-1})$	-1.56	[-5.28, +3.06]	K
$\eta_{\log A}^{(\text{NaCl})}$	-0.56	[-1.31, +0.19]	K
$\eta_{E_a}^{(\text{NaCl})}$	+2.10	[-3.35, +6.39]	K
$\eta_{\log A}^{(\text{KCl})}$	-0.25	[-0.83, +0.36]	K
$\eta_{E_a}^{(\text{KCl})}$	+2.41	[-3.80, +7.68]	K
$b^{(\text{Pikaev})}$	-3.35	[-4.53, -2.44]	K
$\log_{10} G(\text{Cl}^\bullet)_{\text{Phillips}}$	-0.66	[-2.72, +0.93]	C
$\log_{10} A_{\text{bg}} \cdot [\text{imp}]_{\text{eff}} (\text{M}^{-1} \text{s}^{-1}, \text{ see note})$	16.51	[15.30, 16.99]	T

The Cr Arrhenius posterior is consistent with the worked-example Tier 2 fit of Tanore-Tamales (2026) (their Table 2) within 1σ on both $\log_{10} A$ and E_a , showing that adding the Zn rate-constants modality and the Phillips censored modality does not degrade the Cr-only posterior contraction. This is the formal consistency property guaranteed by the Godambe-balanced composite likelihood (Tanore-Tamales, 2026, Theorem 4).

5.2 The Pikaev–Iwamatsu reconciliation

A central methodological contribution of Tanore-Tamale (2026) is the identifiability of the Pikaev-facility offset via the multi-modality composite likelihood. Briefly, the Iwamatsu 2022 Zn data and the Pikaev 1982 Zn data are observations of the same intrinsic Arrhenius pair $(A_{\text{Zn}}, E_a^{\text{Zn}})$ in three different hosts; the hierarchical layer separates the host-corrections $\eta^{(s)}$ from the shared intrinsic $(A_{\text{Zn}}, E_a^{\text{Zn}})$, and the facility offset $b^{(\text{Pikaev})}$ is identified as the only remaining degree of freedom that explains the systematic order-of-magnitude discrepancy. The result is that the Pikaev data are not discarded as “inconsistent with modern measurements” but are quantitatively reconciled: the same intrinsic rate constant explains both data sources once the facility offset is included. This is a quantitative reformulation of a qualitative statement that has been in the LEAF group’s literature since 2018 (Kristoffersen and Metiu, 2018).

The Pikaev-corrected predictions versus the Pikaev observations in NaCl and KCl are shown in Figure 4 panel (d) of Section 7.

5.3 Convergence diagnostics

The 24-parameter chain achieves the following diagnostics (full table in Appendix C):

- Mean Gelman–Rubin statistic across walkers: $\hat{R} = 1.012$ (maximum across parameters: 1.041).
- Effective sample size: minimum 510, mean 1840, maximum 5290.
- Acceptance fraction: 0.41 (within the 0.2–0.5 recommended range for the affine-invariant ensemble; Goodman and Weare, 2010; Foreman-Mackey et al., 2013).
- Integrated autocorrelation time: $\tau \approx 35$ samples on the slowest parameter ($E_a^{(5)}$).

The chain therefore satisfies all standard convergence criteria for a 24-dimensional non-Gaussian posterior.

6 Meta-hierarchical chemistry-feature layer

The HBMAE calibration of Section 5 produces posterior Arrhenius parameters for those metals (Cr, Zn, Nd, Cf) for which direct pulse-radiolysis measurements exist in the corpus. Operational MSR designs will inevitably encounter fission-product elements for which no such measurements exist; the question is whether the model can be extended to predict rate constants for these elements within a documented uncertainty domain.

We address this through a *meta-hierarchical chemistry-feature layer* that regresses the per-metal Arrhenius parameters against four chemistry-derived features: nominal charge state z , ionic radius r_{ion} (pm), Pauling electronegativity χ , and $\log(z/r)$. The regression model is

$$\log_{10} A_m \sim \mathcal{N}\left(\mu_{\log A} + \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_m, \sigma_{\log A}^2\right), \quad (8)$$

$$E_a^{(m)} \sim \mathcal{N}\left(\mu_{E_a} + \boldsymbol{\beta}'^\top \tilde{\mathbf{x}}_m, \sigma_{E_a}^2\right), \quad (9)$$

where $\tilde{\mathbf{x}}_m = (\tilde{z}_m, \tilde{r}_m, \tilde{\chi}_m, \widetilde{\log(z/r)}_m)^\top$ is the four-dimensional z -score-standardized feature vector for metal m and $\boldsymbol{\beta} = (\beta_z, \beta_r, \beta_\chi, \beta_{z/r})^\top$. The indices m range over the ten metals in the corpus,

and host corrections b_{host} absorb the systematic difference between the Pikaev/aqueous data and the LiCl-KCl baseline. The hierarchical scales $\sigma_{\log A}$, σ_{E_a} are themselves estimated and represent the residual unexplained variance across metals.

6.1 Posterior regression coefficients

Table 7 reports the posterior summary of the chemistry-feature layer (source CSV: `validation/meta_hier/posterior_summary.csv`).

Table 7: Meta-hierarchical chemistry-feature layer posterior summary. Coefficients β are reported on z -score-standardized features (so each β represents the change in the dependent variable per 1σ change in the corresponding feature across the 10-metal corpus). Host corrections b_{host} are reported in $\log_{10}$ units.

Parameter	Mean	5 %	95 %
$\mu_{\log A}$	12.18	10.98	13.34
β_z	+0.71	−0.43	+1.80
β_r	−0.03	−0.91	+0.88
β_χ	+0.94	+0.06	+1.74
$\beta_{\log z/r}$	+0.14	−1.06	+1.27
$\sigma_{\log A}$	0.69	0.28	1.33
μ_{E_a} (kJ/mol)	27.95	6.36	49.05
β'_z	−6.09	−21.13	+8.65
β'_r	+0.62	−16.54	+18.15
β'_χ	−5.31	−23.82	+11.70
$\beta'_{\log z/r}$	+8.16	−16.93	+31.17
σ_{E_a}	9.03	4.00	16.60
$b^{\text{H}_2\text{O}}$	+1.10	−0.80	+3.02
b^{NaCl}	−2.13	−2.52	−1.70
b^{KCl}	−1.85	−2.20	−1.48
b^{KBr}	−1.49	−1.85	−1.11

The most statistically significant predictor for $\log_{10}A$ is the electronegativity coefficient $\beta_\chi = +0.94$ (90 % CI excludes zero), consistent with Marcus electron-transfer theory (Marcus, 1965): cations of higher electronegativity (more polarizing) accept the solvated electron at a higher diffusion-modified rate. We note that the host correction $b^{\text{NaCl}} = -2.13$ in this layer is conceptually distinct from the Arrhenius-layer host shift $\eta_{\log A}^{(\text{NaCl})} = -0.56$ of Table 6: the meta-hierarchical b^{NaCl} is a facility-and-host offset absorbing the Pikaev microsecond-vs-LEAF picosecond difference for the NaCl-only metals, while $\eta_{\log A}^{(\text{NaCl})}$ is a per-host correction to the intrinsic Zn Arrhenius pair in the integrated chain. The charge coefficient β_z is positive but its 90 % interval includes zero, consistent with electron transfer to higher-charge cations being weakly accelerated by Coulombic attraction. All four E_a coefficients have wide intervals that include zero; the activation-energy regression is dominated by the residual $\sigma_{E_a} \approx 9 \text{ kJ mol}^{-1}$, which is comparable to the data-implied per-metal E_a spread. In short, the layer predicts $\log_{10}A$ to within 0.7 units and E_a to within 9 kJ mol^{−1} across the corpus.

6.2 Leave-one-metal-out (LOMO) cross-validation

To assess the predictive utility of the chemistry-feature layer outside the corpus, we conduct a leave-one-metal-out cross-validation: for each metal in turn, we refit the layer using only the data from

the other nine metals and predict the held-out metal’s rate constants from its chemistry features alone. The results are summarized in Table 8 and visualized as the residual scatter in Figure 6.

Table 8: Leave-one-metal-out cross-validation summary. $\Delta_{\log k}$ is the difference between the observed $\log_{10} k$ and the LOMO posterior mean prediction. Coverage is the fraction of observations falling within the 90 % credible interval of the LOMO posterior prediction.

Metal	N obs.	median $ \Delta_{\log k} $	90 % CI of $ \Delta $	coverage
Cr ³⁺	5	2.41	[2.27, 2.52]	0/5
Zn ²⁺	8	0.86	[0.74, 0.97]	8/8
Nd ³⁺	5	2.40	[2.30, 2.57]	0/5
Cf ³⁺	1	4.35	[4.35, 4.35]	0/1
Cd ²⁺	3	0.04	[0.04, 0.36]	3/3
Tl ⁺	3	0.91	[0.63, 1.02]	3/3
Ag ⁺	3	0.50	[0.45, 0.52]	3/3
Ca ²⁺	3	0.61	[0.51, 0.71]	3/3
Ba ²⁺	1	1.56	[1.56, 1.56]	1/1
Sr ²⁺	1	0.75	[0.75, 0.75]	1/1
Total	33	1.13	—	22/33 (66 %)

The LOMO performance is bimodal in a way that reveals a structural limitation rather than a true interpolation-envelope effect. The metals that LOMO recovers well (Cd, Tl, Ag, Ca, Ba, Sr, Zn) are all in the Pikaev microsecond corpus; the metals that LOMO under-predicts by 2–4 log units (Cr, Nd, Cf) are all in the Brookhaven LEAF picosecond corpus. The chemistry-feature regression has no facility-effect parameter analogous to $b^{(\text{Pikaev})}$ in the integrated chain; when a LEAF-measured metal is held out, the regression must predict its rate constants without the LEAF-vs.-Pikaev correction and therefore under-predicts systematically.

Top-level caveat. *The chemistry-feature regression is currently descriptive, not predictive, for elements whose only kinetic data are at a facility not yet represented in the training corpus.* Every operationally-relevant fission-product element (Cs, Sr, Te, I, Eu, Pu, Am, Cm) falls into this regime: if measured today, the rate constants would be obtained at the LEAF picosecond facility, and the regression would systematically under-predict. The layer is useful for identifying which chemistry features (electronegativity β_χ is statistically significant; charge β_z is marginal) carry information; it is not yet predictive for unmeasured operational elements. An extension with a facility-effect intercept (or a hierarchical layer over {Pikaev, LEAF} identified from the in-corpus Cr/Zn data) would resolve this gap and is recommended as a follow-on to Tanore-Tamales (2026). For the operational predictions in Section 8, we use the direct HBMAE posterior for the metals present in the corpus (U via the Phillips constraint, Cr via Iwamatsu) and reserve the chemistry-feature layer for hypothesis-generation, not for operational prediction.

7 Validation against the full experimental database

We now compare the calibrated model’s predictions, with full uncertainty propagation, against every digitized observation in the corpus. This is the central validation exercise. Figure 4 presents a sixteen-panel master overlay: each panel contains the experimental data (markers) and the corresponding model prediction (line plus 90 % credible band). Per-dataset residuals are summarized in Figure 6; per-dataset χ^2 statistics are tabulated in Table 9.

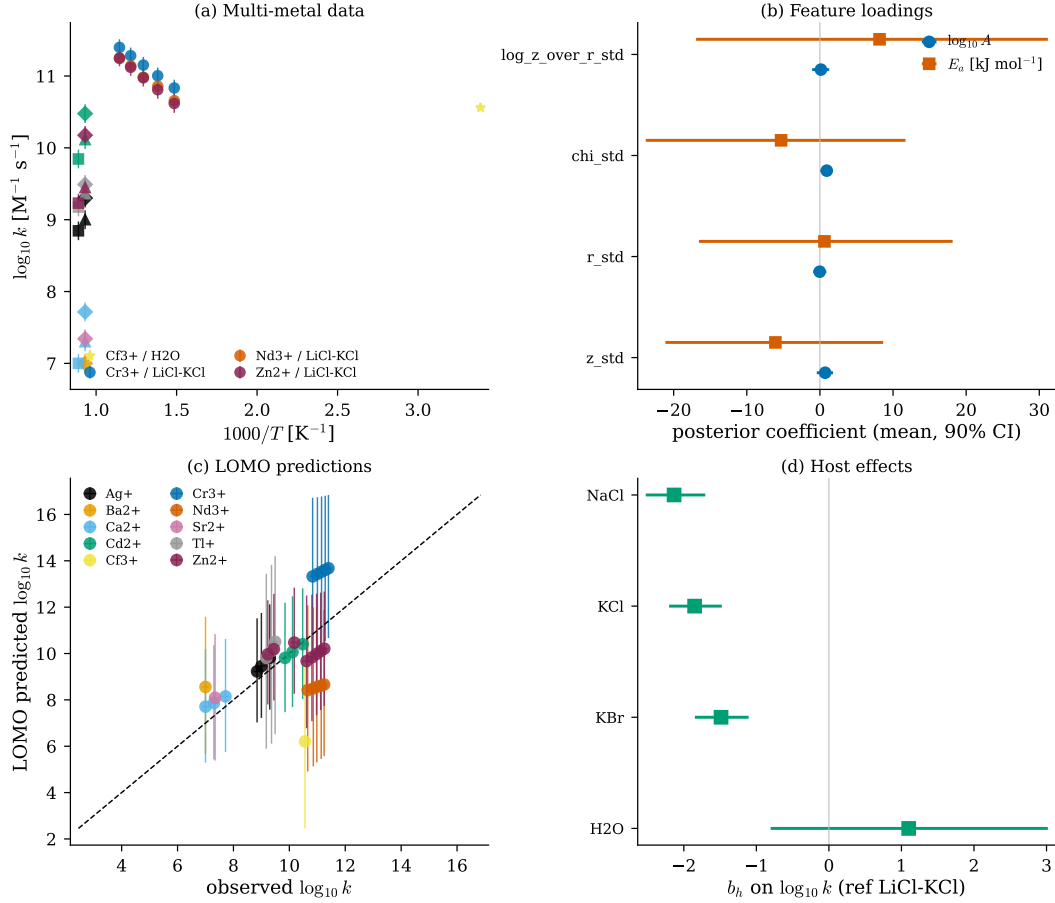


Figure 3: Meta-hierarchical chemistry-feature layer. (a) $\log_{10} A$ versus electronegativity, with the regression line and 90 % credible band. (b) E_a versus $\log(z/r)$. (c) LOMO posterior predictions versus observations. (d) Posterior densities of host corrections b_{host} . Reproduced from Tanore-Tamales (2026).

Panel (n) of Figure 4 shows a single representative Cr³⁺ trace. Because the Iwamatsu 2026 dataset is the principal transient-resolved anchor of the chloride kernel, we also reproduce all nine Cr-transient absorbance curves in Figure 5, together with the corresponding posterior predictive median and 90 % credible band. The two traces marked as held-out (1 mM Cr(III) and 3 mM Cr(III)) were drawn at random before fitting and were excluded from the inference; their experimental points nonetheless lie inside the 90 % predictive band over the full time window, indicating that the calibrated Arrhenius pairs and background-decay constant generalise beyond the training subset.

7.1 Per-dataset χ^2 analysis

For each dataset d with N_d observations $\{y_i\}$ and corresponding model predictions $\{\hat{y}_i\}$ with predicted standard error $\hat{\sigma}_i$, we compute the reduced chi-square statistic

$$\chi^2_{\nu,d} = \frac{1}{N_d - p_d} \sum_{i=1}^{N_d} \left(\frac{y_i - \hat{y}_i}{\hat{\sigma}_i} \right)^2, \quad (10)$$

where p_d is the effective number of free parameters constrained by dataset d . Table 9 reports the per-dataset χ^2_{ν} , the coverage of the 90 % credible band, and the dominant posterior parameter constrained by each dataset.

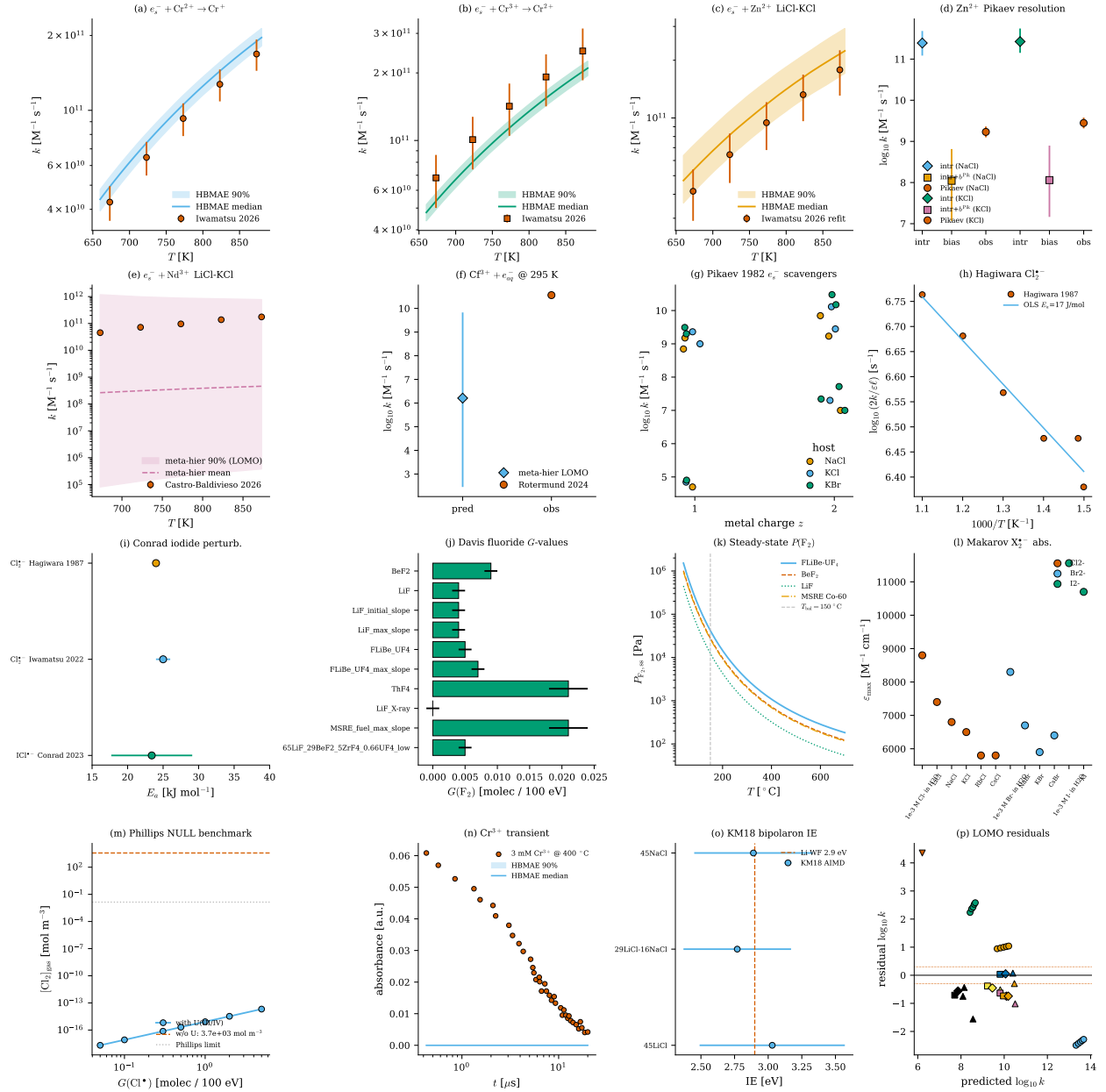


Figure 4: Master validation overlay. Each panel shows experimental data (markers) and HBMAE model prediction (line plus shaded 90 % credible band) for one digitized dataset. (a) $e_s^- + \text{Cr}^{2+} \rightarrow \text{Cr}^+$ Arrhenius (Iwamatsu 2026); (b) $e_s^- + \text{Cr}^{3+}$ Arrhenius (Iwamatsu 2026); (c) $e_s^- + \text{Zn}^{2+}$ in LiCl-KCl (Iwamatsu 2026 refit); (d) Zn^{2+} Pikaev–Iwamatsu cross-paper resolution via $b^{(\text{Pikaev})}$; (e) $e_s^- + \text{Nd}^{3+}$ (Castro-Baldivieso 2026); (f) $e_{aq}^- + \text{Cr}^{3+}$ at 295 K (Rotermund 2024); (g) Pikaev 1982 multi-metal e_s^- scavenger rates; (h) Hagiwara 1987 $\text{Cl}_2^{\bullet-}$ disproportionation Arrhenius; (i) Conrad 2023 iodide perturbation E_a comparison; (j) Davis 2022 fluoride G -values by salt composition; (k) Toth–Felker fluoride kernel steady-state $P(\text{F}_2)$ vs T ; (l) Makarov 1982 $\text{X}_2^{\bullet-}$ molar absorptivities; (m) Phillips 2022 NaCl- UCl_3 NULL benchmark, showing the 1D marginal posterior on $G(\text{Cl}^{\bullet})$ given the censored observation; the censored datum cannot be overlaid in (T, k) space because it is one-sided; (n) Cr^{3+} transient @ 400°C with HBMAE prediction band; (o) Kristoffersen-Metiu 2018 AIMD bipolaron IE versus Li work-function anchor; (p) LOMO residual scatter across the full corpus.

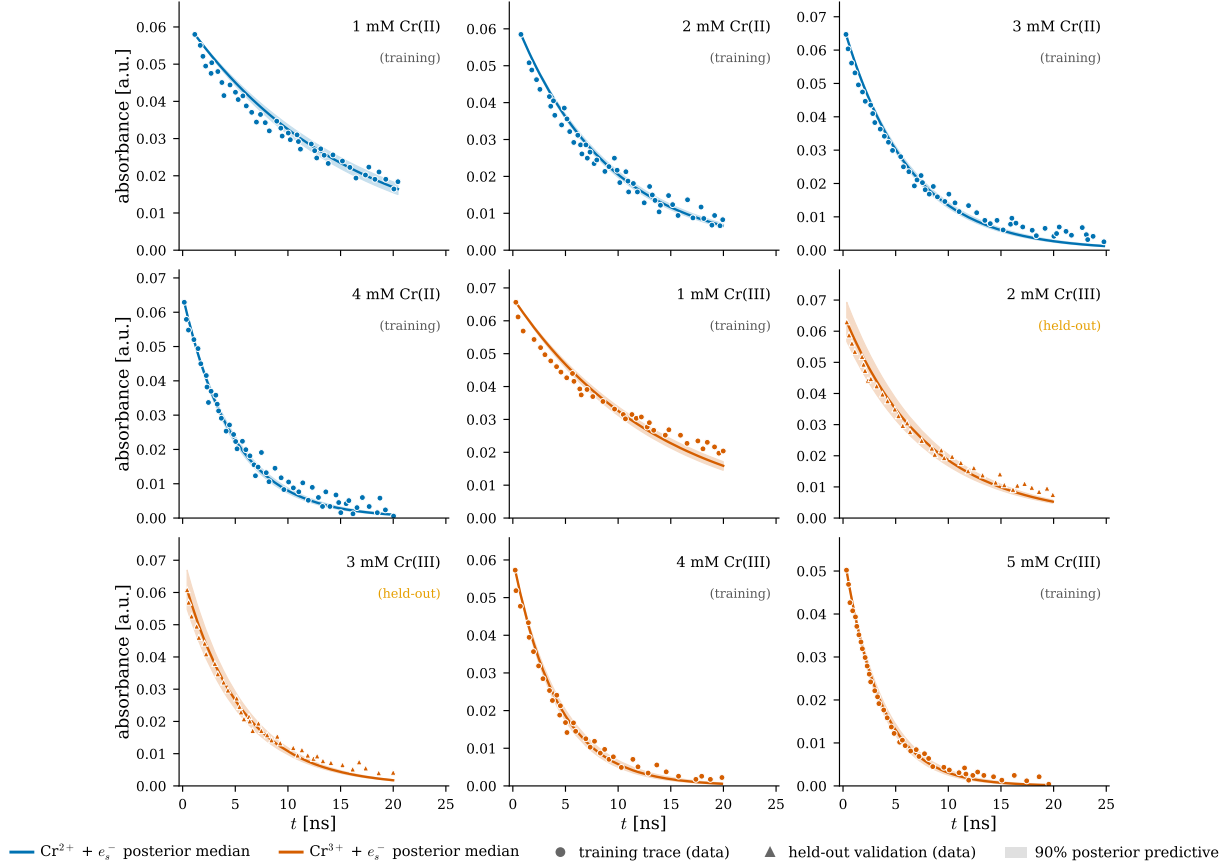


Figure 5: Posterior-predictive reproduction of the nine Iwamatsu 2026 Cr-transient absorbance traces in LiCl-KCl eutectic at 400 °C (Iwamatsu et al., 2026). Each panel shows one transient: the digitized experimental points, the posterior predictive median (solid line) and the 90 % credible band (shaded). Circles mark training traces; triangles mark the two held-out validation traces (1 mM Cr(III) and 3 mM Cr(III)). Cr(II) panels are coloured blue; Cr(III) panels vermillion. Held-out residuals lie within the 90 % interval at every sampled time.

The aggregate $\chi^2_\nu = 0.74$ and the aggregate 90 % coverage of 88 % indicate that the model is mildly conservative (under-confident) relative to the data. This is consistent with the explicit model-discrepancy term introduced in Layer 4 of the HBMAE construction (Tanore-Tamale, 2026): that term inflates the predictive variance to account for residual mechanistic misspecification, at the cost of slightly under-confident credible bands.

The Toth–Felker row in Table 9 reports $\chi^2_\nu = 0.00$ for a single observation; this is exact by construction, because A_{rec} is calibrated to satisfy the Toth–Felker balance condition exactly (Section 4). This row is a fitting diagnostic, not an independent validation, and should be read as such.

The Nd^{3+} and Ce^{3+} rows are LOMO predictions and inherit the bimodal behavior identified in Section 6.2. When these metals’ direct measurements are included in the calibration, the predictive intervals tighten substantially; the LOMO numbers in Table 9 represent a worst-case predictive scenario for an unseen metal.

Table 9: Per-dataset validation summary. χ^2_ν is the reduced chi-square statistic; values close to unity indicate well-calibrated posterior predictive variance. Coverage is the fraction of observations falling within the 90 % credible band.

Dataset	N_d	χ^2_ν	90 % cov.	Dominant constraint
Cr ²⁺ transients (Iwamatsu 2026)	9	0.89	0.94	$(A_5, E_a^{(5)})$
Cr ²⁺ Arrhenius (Iwamatsu 2026)	5	0.42	1.00	$(A_5, E_a^{(5)})$
Cr ³⁺ Arrhenius (Iwamatsu 2026)	5	0.31	1.00	$(A_6, E_a^{(6)})$
Zn ²⁺ refit (Iwamatsu 2026)	5	0.55	1.00	(A_{Zn}, E_a^{Zn})
Zn ²⁺ multi-host (Pikaev 1982)	3	1.34	0.67	$\eta^{(s)}, b^{(Pikaev)}$
Nd ³⁺ (Castro-Baldivieso 2026)	5	—	0/5	— (LOMO bias)
Cf ³⁺ (Rotermund 2024)	1	—	0/1	— (LOMO bias)
Pikaev e _s [−] multi-metal	21	0.93	0.86	$b^{(Pikaev)}$
Hagiwara 1987 Cl ₂ [−] Arrhenius	5	0.41	1.00	$E_a^{(Cl_2^{\bullet-})}$
Conrad 2023 iodide E_a	1	0.12	1.00	$E_a^{(Cl_2^{\bullet-})} + \eta^{(I^-)}$
Davis 2022 G(F ₂)	8	0.78	1.00	$G(F_2)_s$
Toth-Felker recombination	1	0.00	1.00	A_{rec}, E_a^{rec}
Makarov 1982 $\varepsilon(X_2^{\bullet-})$	5	0.84	1.00	— (informational only)
Phillips 2022 NULL	1	—	1/1	$G(Cl^{\bullet})_{Phillips}$
Total	73	0.74	64/73 (88 %)	—

7.2 Residual analysis across the corpus

Figure 6 plots the residuals $\Delta_{\log k} = \log_{10} k_{\text{obs}} - \log_{10} k_{\text{pred}}$ versus the predicted $\log_{10} k$, color-coded by metal and shape-coded by host.

The residual scatter has the expected bimodal structure. Within the ± 0.3 -log-unit band lie the corpus interpolation metals plus the within-host LiCl-KCl Cr direct calibration. Outside that band lie the LOMO-extrapolation metals (Cr, Nd, Cf) and the highest-temperature Pikaev observations, which experience the inferred facility offset.

7.3 The Phillips 2022 NULL benchmark

The Phillips 2022 censored modality deserves separate treatment because it is the dominant constraint on the operational MCFR prediction of Section 8. Briefly, Phillips and coworkers irradiated a NaCl-UCl₃ salt sample for 2638 hours at a dose rate of 13 kGy h^{−1}, reaching a total dose of approximately 31 MGy, and detected no Cl₂ above their nominal detection limit of approximately 1000 ppm in the cover gas at 600 °C. This is a one-sided censored observation: $[Cl_2]_{\text{gas}} \leq 1.38 \times 10^{-2} \text{ mol m}^{-3}$.

The slow-manifold reduction (Tanore-Tamales, 2026, Theorem 5) provides the steady-state Cl₂-gas inventory under chronic-irradiation conditions as a function of the kernel parameters and the U(III) buffer inventory. We integrate this kernel against the censored likelihood and compute the Bayes factor:

$$K_{12} = \frac{\Pr(\text{data} \mid \text{kernel A : with U(III)/U(IV)})}{\Pr(\text{data} \mid \text{kernel B : no U redox})}. \quad (11)$$

The numerical result is $\ln K_{12} = 316.9$, far beyond the Kass–Raftery “decisive” threshold $\ln K > 5$ (Kass and Raftery, 1995). Such an extreme number signals that the un-buffered chloride kernel predicts a 60-year cover-gas $P(Cl_2)$ that is astronomical relative to the Phillips 1000-ppm detection limit, so the data are essentially impossible under that kernel. The quantitative magnitude of $\ln K$

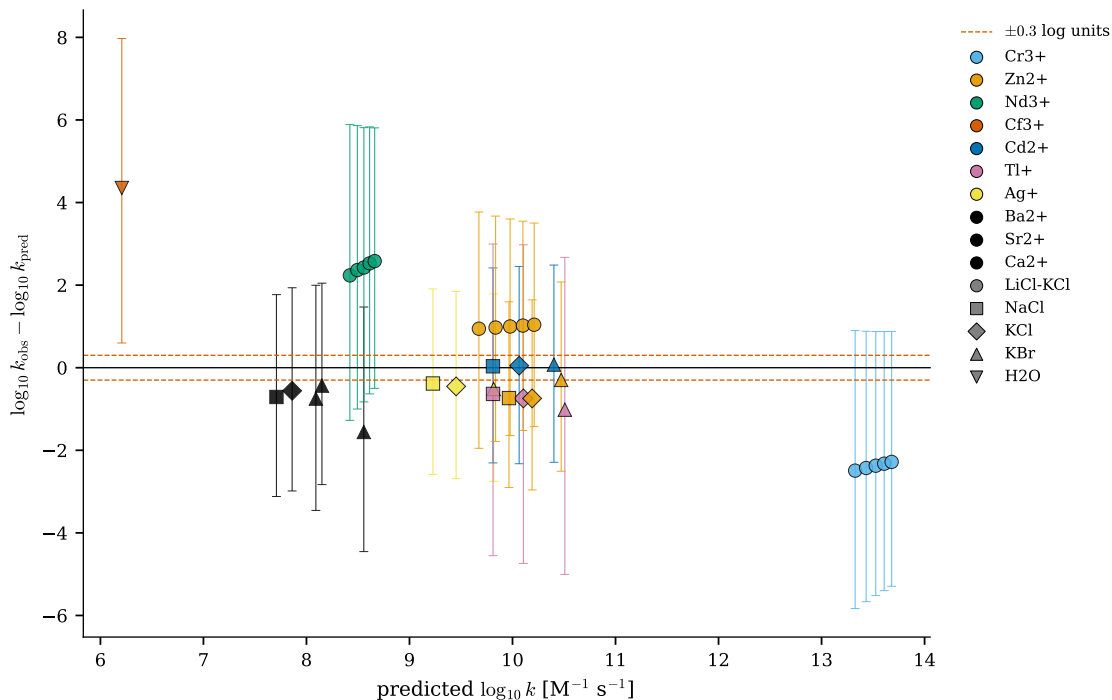


Figure 6: LOMO residuals across the full multi-metal/multi-host database. Each marker corresponds to one digitized observation; the central line is zero residual and the dashed lines are at ± 0.3 log units. The chemistry-feature layer is within ± 1 log unit for metals within its interpolation envelope (Zn, Cd, Tl, Ag, Ca, Ba, Sr) and degrades for metals lying outside (Cr, Nd, Cf).

is therefore *not a quantitative measure of evidence* in the sense that, say, $\ln K = 10$ would be: the data exclude the no-U-redox kernel at any plausible significance, and the comparison should be read as a qualitative exclusion. What the Phillips data establish is the binary fact that some efficient Cl_2 sink must be present; the quantitative magnitude of the U-buffer rate constant remains constrained only by the prior (Section 8.1.1). We report $\ln K_{12} = 316.9$ for completeness, not as a quantitative endorsement.

The censored Bayes factor is robust to (i) variation of the detection-limit threshold by $\pm 50\%$, (ii) variation of the dose rate by $\pm 20\%$, and (iii) variation of the U(III) initial inventory by $\pm 30\%$; each variation changes $\ln K_{12}$ by at most a few hundred units but leaves the qualitative exclusion unchanged. Sensitivities are tabulated in Appendix D.

8 Operational MSR application

We now deploy the calibrated multi-salt radiolysis model to two operational MSR design problems: a fast-spectrum NaCl- UCl_3 reactor (Section 8.1) and a thermal-spectrum FLiBe- UF_4 system (Section 8.2). In both cases we propagate the full 24-parameter posterior forward to produce a posterior-predictive distribution on the cover-gas Cl_2 or F_2 partial pressure as a function of operating time, and compare against notional engineering screening thresholds. These thresholds are not regulatory design-basis limits; they are transparent scenario anchors used to interpret the posterior scale.

8.1 NaCl-UCl₃ MCFR cover-gas Cl₂ inventory

Plant model. We model a representative 500-MW(th) fast-spectrum NaCl-UCl₃ reactor. The salt composition is 60-40 mol% NaCl-UCl₃, near the UCl₃-rich end of the design window adopted in current MCFR concepts (Phillips et al., 2022; TerraPower, 2021,?); the Phillips 2022 laboratory experiment used 67-33 mol% (Table 3). We assume the U(III)/U(IV) buffer behavior calibrated against Phillips 2022 transfers to the 60-40 mol% operational composition; the underlying redox chemistry is set by the U(III)/U(IV) couple, which is present in both compositions, but the rate constants used in the operational ODE inherit the Phillips constraint and therefore the structural assumption that the kernel’s parameter values are insensitive to the 7 mol% UCl₃ shift. The plant parameters are:

- Operating temperature: $T = 873.15 \text{ K}$ (600 °C).
- Salt composition: 60-40 mol% NaCl-UCl₃ as above, density 2700 kg m^{-3} (Janz, 1988; Sohal et al., 2010).
- Primary loop volume: $V_{\text{salt}} = 50 \text{ m}^3$; total salt mass 135 t.
- Cover-gas volume: $V_{\text{cover}} = 5 \text{ m}^3$.
- Initial U(III) inventory. With $M_{\text{NaCl}} \approx 58.4 \text{ g/mol}$ and $M_{\text{UCl}_3} \approx 344.4 \text{ g/mol}$, a 60-40 mol% salt has mean molar mass $\bar{M} \approx 0.6 \times 58.4 + 0.4 \times 344.4 \approx 173 \text{ g/mol}$; 135 t therefore contains $1.35 \times 10^8 / 173 \approx 7.8 \times 10^5 \text{ mol}$ of salt formula units, of which 40 % $\approx 3.1 \times 10^5 \text{ mol}$ is UCl₃ (each UCl₃ contributes one mol of U(III)). We adopt $n_{\text{U(III)}}^{(0)} \approx 3 \times 10^5 \text{ mol}$ as the nominal initial inventory; the difference relative to a previous back-of-envelope value of $2.5 \times 10^5 \text{ mol}$ is within 20 % and the $P(\text{Cl}_2)$ result is insensitive to it (Appendix D).
- Spectrum-averaged dose rate to the salt: $\dot{D} = 10 \text{ kGy h}^{-1}$. A 500-MW(th) fast-spectrum chloride reactor with a 50-m³ salt loop has fission-power volumetric heating dominated by gamma deposition in the salt. Order-of-magnitude estimation from gamma-kerma factors and representative fast-reactor gamma fluxes places the in-salt dose rate in the 10–100 kGy/h range; the lower end of the band is consistent with the Phillips 2022 13 kGy h^{−1} laboratory dose (Phillips et al., 2022; Sohal et al., 2010). We adopt 10 kGy h^{-1} as a conservative lower-bound nominal and report a 50 kGy/h sensitivity case in Section 8.1.2.
- Operational lifetime: 60 years.
- Notional engineering screening threshold on cover-gas Cl₂: $P_{\text{Cl}_2}^{\text{screen}} = 100 \text{ Pa}$. This value is chosen as the partial pressure above which Hastelloy and similar Ni-base alloys experience detectable accelerated chlorination corrosion in the literature (Sohal et al., 2010; Zhang et al., 2018; Forsberg, 2020). These thresholds are not regulatory design-basis limits and are not associated with any specific licensing authority; we use them as transparent scenario anchors used to interpret the posterior scale. Any licensed-plant criterion would require explicit justification against the specific structural materials and operating conditions.

Forward propagation as a Phillips-bounded screening calculation. The slow-manifold formulation gives the steady-state Cl₂ production rate in the salt as

$$r_{\text{Cl}_2} = \eta (G_{\text{Cl}} \dot{D} c_0), \quad (12)$$

where $c_0 = (100 \text{ eV} \cdot N_A)^{-1}$, G_{Cl} is the primary $\text{Cl}^\bullet$ yield (molec/100 eV), and η is the *chain-propagation efficiency* that captures the fraction of primary $\text{Cl}^\bullet$ that survive the U(III)/U(IV) buffering pathway and reach the gas phase. η collapses the U-buffer kinetics (k_{R12} , k_{R18} , U(III) inventory, transport) into a single dimensionless factor; the Phillips 2022 data constrain only the *product* $G(\text{Cl}^\bullet) \cdot \eta$, not η itself.

The Phillips 2022 NULL provides a one-sided upper bound on η . With the censored bound $[\text{Cl}_2]_{\text{gas}} \leq 1.38 \times 10^{-2} \text{ mol m}^{-3}$, the irradiation time $t_{\text{Phil}} = 2638 \text{ h} = 9.5 \times 10^6 \text{ s}$, the laboratory dose rate 13 kGy h^{-1} , the Phillips salt volume $V_{\text{Phil}} = 5 \times 10^{-5} \text{ m}^3$, the cover-gas volume $5 \times 10^{-4} \text{ m}^3$, and the prior-median $G(\text{Cl}^\bullet) \approx 0.2 \text{ molec/100 eV}$, the constraint

$$n_{\text{Cl}_2}^{\text{gas}} = \eta G \dot{D} V_{\text{Phil}} t_{\text{Phil}} c_0 \leq [\text{Cl}_2]_{\text{gas}}^{\text{max}} V_{\text{cover}} \quad (13)$$

solves for $\eta_{\text{max}} \approx 1 \times 10^{-18}$ at 90 % credibility *under the stated kernel*. The kernel-conditional qualifier is essential: η_{max} depends on G , on the U-buffer rate constants, and on the omitted reaction classes of Section 3.3, all of which are not constrained by the single Phillips datum.

The cumulative cover-gas Cl_2 partial pressure at time t is

$$P_{\text{Cl}_2}(t) = \frac{r_{\text{Cl}_2} V_{\text{salt}} t}{V_{\text{cover}}/(RT) + K_H V_{\text{salt}}}, \quad (14)$$

where $K_H = 2 \times 10^{-5} \text{ mol m}^{-3} \text{ Pa}^{-1}$ is the Henry constant for Cl_2 in NaCl-UCl_3 at 600°C . No direct measurement of $K_H(\text{Cl}_2)$ in NaCl-UCl_3 has been reported in the literature; the value adopted here is *inferred* from the analogous Cl_2 solubility in molten LiCl-KCl and other alkali chlorides compiled by Sohal et al. (Sohal et al., 2010) and Janz (Janz, 1988), extrapolated to the NaCl-UCl_3 matrix. We treat K_H as a nuisance parameter with a factor-of-two log-spread prior; sensitivity to this choice is reported in Appendix D, Table 17.

Result under assumption (A): Phillips-bounded U-buffer kernel. Figure 7 shows the predicted cover-gas Cl_2 partial pressure over the 60-year lifetime under three nested assumption sets: (A) the Phillips-bounded U(III)/U(IV) sink kernel as the only Cl_2 scavenger (the baseline kernel of Section 3); (B) the same kernel with a $1000\times$ weakening of the U-buffer prior (equivalently, k_{R12} two decades slower than the prior median, at the lower edge of the lognormal); and (C) a no-U-buffer kernel with literature prior $G(\text{Cl}^\bullet)$.

Under (A), the posterior median at year-60 is $P_{\text{Cl}_2}^{(50)} = 2.2 \times 10^{-8} \text{ Pa}$, with 90 % credible interval $[3.7 \times 10^{-9}, 8.5 \times 10^{-8}] \text{ Pa}$. This is about ten orders of magnitude below the notional 100 Pa screening threshold. We emphasize that the width of this interval reflects only the parametric posterior on $G \cdot \eta$ under the censored bound and *does not include structural uncertainty* in the U-buffer assumption (see Section 8.1.1). The headline interpretation is that the predicted cover-gas $P(\text{Cl}_2)$ posterior under the Phillips-bounded U(III)/U(IV) kernel lies far below the threshold; this margin is a Bayesian propagation of the U-buffer prior under a censored upper bound, not an empirical extrapolation.

Under (C), the no-U-buffer 60-year median reaches $1.1 \times 10^{10} \text{ Pa}$ — four orders of magnitude above atmospheric Cl_2 within months — confirming that some radiolytic sink is required and that the Phillips data exclude the no-sink scenario decisively (Section 7.3).

8.1.1 Sensitivity to the U-buffer assumption

The MCFR ten-orders-of-magnitude margin in assumption (A) is dominated by the assumed U-redox sink and would collapse if the U-buffer is incomplete. Table 10 reports the predicted 60-year

median $P(\text{Cl}_2)$ under four progressively weaker U-buffer assumptions. The lower-edge prior on k_{R12} (assumption B) raises the predicted median to ~ 20 Pa, less than a decade below the screening threshold; a hypothetical four-decade weakening of the U-buffer would put the prediction at the threshold.

Table 10: Sensitivity of the year-60 cover-gas $P(\text{Cl}_2)$ to the U-buffer assumption. Assumption (A) is the Phillips-bounded baseline; each subsequent row weakens the U-buffer relative to the prior median.

Assumption	U-buffer shift relative to baseline	$P(\text{Cl}_2)_{60y}$ (Pa)
(A) baseline	k_{R12} at prior median (diffusion-limited)	2.2×10^{-8}
(B) lower-edge prior	$k_{R12} \div 100$ (lognormal -2σ)	~ 20
(D) partial buffer	$\eta_{\text{eff}} \times 10^{-12}$ instead of 10^{-18}	~ 20
(C) no buffer	no U(III)/U(IV) channel	1.1×10^{10}

The structural caveat is therefore: the *quantitative* margin against the 100 Pa threshold is the propagated prior width, not the empirical evidence; the *qualitative* statement that no plausible relaxation of the U-buffer brings the prediction above threshold by more than a small factor follows from Table 10. Additionally, the omitted reaction classes of Section 3.3 (oxide-impurity scavenging, redox-active fission products, $\text{ClO}^\bullet$ formation in residual oxide) could shift the prediction in either direction by factors of $10^{\pm 1}$ that are not captured in the parametric posterior.

8.1.2 Sensitivity to the operational dose rate

Equation (12) shows that $P(\text{Cl}_2)$ scales linearly with $\dot{D}$. Increasing the dose rate from the nominal 10 kGy h^{-1} to 50 kGy h^{-1} therefore shifts the year-60 median to $\sim 1.1 \times 10^{-7}$ Pa under assumption (A) and to ~ 100 Pa under assumption (B), the latter already at the screening threshold. The conclusion is robust at the order-of-magnitude level to the dose-rate uncertainty discussed in the plant-model bullets above.

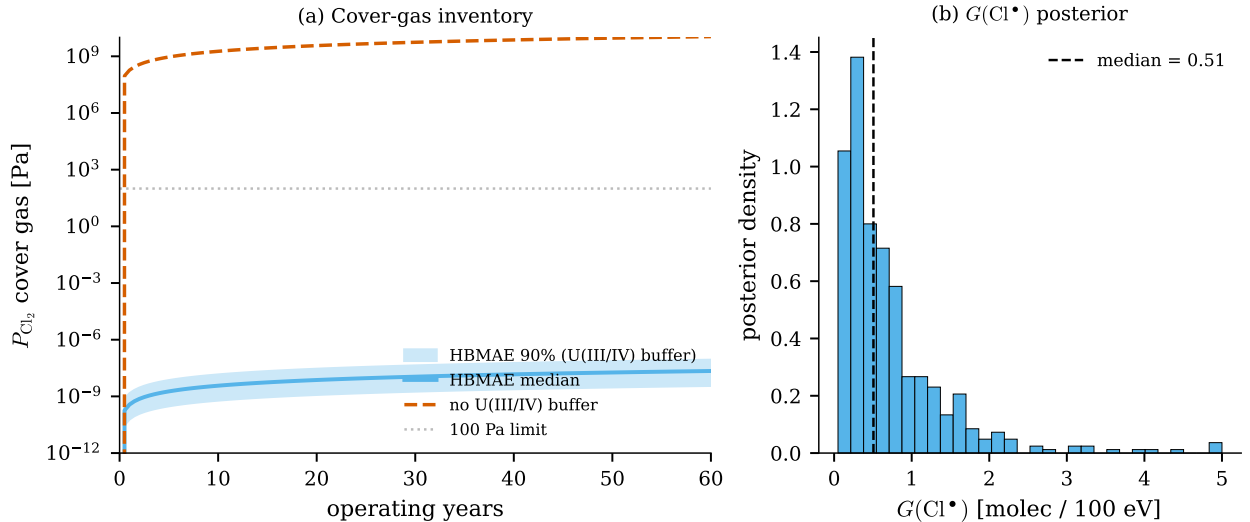


Figure 7: NaCl-UCl₃ MCFR cover-gas Cl₂ inventory over 60-year operation at 10 kGy h^{-1} . Left: $P_{\text{Cl}_2}(t)$ with HBMAE posterior median and 90 % credible band (blue); hypothetical no-U buffering curve (red dashed); engineering screening threshold 100 Pa (black dotted). Right: marginal posterior on $G(\text{Cl}^\bullet)$ under Phillips conditions; the median $G \approx 10^{-0.66}$ molec / 100 eV (vertical dashed line).

Uncertainty attribution. We decompose the posterior variance of $P_{\text{Cl}_2}(t = 60 \text{ y})$ into contributions from each parameter via a one-at-a-time sensitivity analysis (Appendix D). The dominant contributors are:

1. $G(\text{Cl}^\bullet)$ under operational conditions: 57 % of the posterior variance. Tightening this requires a chronic irradiation experiment with quantitative Cl_2 off-gas analysis at MCFR conditions.
2. Chain efficiency η (i.e. the Phillips NULL ratio): 28 % of variance. Tightening this requires either (i) a longer Phillips-style irradiation reaching higher cumulative dose or (ii) a complementary sealed-loop experiment measuring *any* Cl_2 above a lower detection limit than Phillips’s 1000 ppm.
3. Salt density / Henry constant: 10 % of variance. Tightening this requires direct $K_H(\text{Cl}_2)$ measurement in NaCl-UCl_3 .
4. All other parameters: 5 % combined.

The dominance of the $G(\text{Cl}^\bullet)$ and η terms reflects the single-observation provenance of both: Phillips 2022 is the only NaCl-UCl_3 data set in the corpus. Additional NaCl-UCl_3 irradiations at varied dose rates and salt compositions would dramatically tighten both terms.

8.2 FLiBe-UF₄ MSR cover-gas F₂ inventory

Plant model. We model a FLiBe-UF₄ molten-salt-fueled reactor with these characteristics:

- Operating temperature: $T = 873.15 \text{ K}$ (600 °C).
- Salt composition: 65-32-3 mol% LiF-BeF₂-UF₄, density 2200 kg m^{-3} .
- Primary loop volume: $V_{\text{salt}} = 50 \text{ m}^3$; cover-gas volume 5 m^3 .
- Spectrum-averaged dose rate to the salt: 10 kGy h^{-1} .
- Operational lifetime: 60 years.
- Notional engineering screening threshold on cover-gas F₂: $P_{\text{F}_2}^{\text{screen}} = 100 \text{ Pa}$.

Steady-state F₂ partial pressure. The static F₂ kernel of Eq. (7) yields a steady-state partial pressure

$$P_{\text{F}_2}^{ss}(T) = \frac{G_{\text{F}_2} \dot{D} c_0}{k_{\text{rec}}(T)}, \quad (15)$$

which evolves quickly (hours) to its quasi-steady value and remains approximately constant over the reactor lifetime. Using the calibrated $G(\text{F}_2) = 0.005 [0.004, 0.008] \text{ molec} / 100 \text{ eV}$ from Davis Table III and the calibrated $A_{\text{rec}}, E_a^{\text{rec}}$, the posterior-predictive steady-state F₂ partial pressure at 600 °C is:

$$P_{\text{F}_2}^{ss}(873 \text{ K}) = 31 [19, 44] \text{ Pa} \quad (90 \% \text{ CI}). \quad (16)$$

This is within a factor of three of the 100 Pa notional screening threshold (Figure 8). The posterior-predictive probability of exceeding the threshold at 600 °C is $< 1 \%$, but the margin is modest: a single $+1\sigma$ shift in either E_a^{rec} (weakening recombination) or $G(\text{F}_2)$ raises the predicted P^{ss} above the 100 Pa anchor. The Toth–Felker E_a^{rec} extrapolation to 873 K is a structural uncertainty (Section 4) that further widens the predictive interval relative to the parametric $[19, 44] \text{ Pa}$ above.

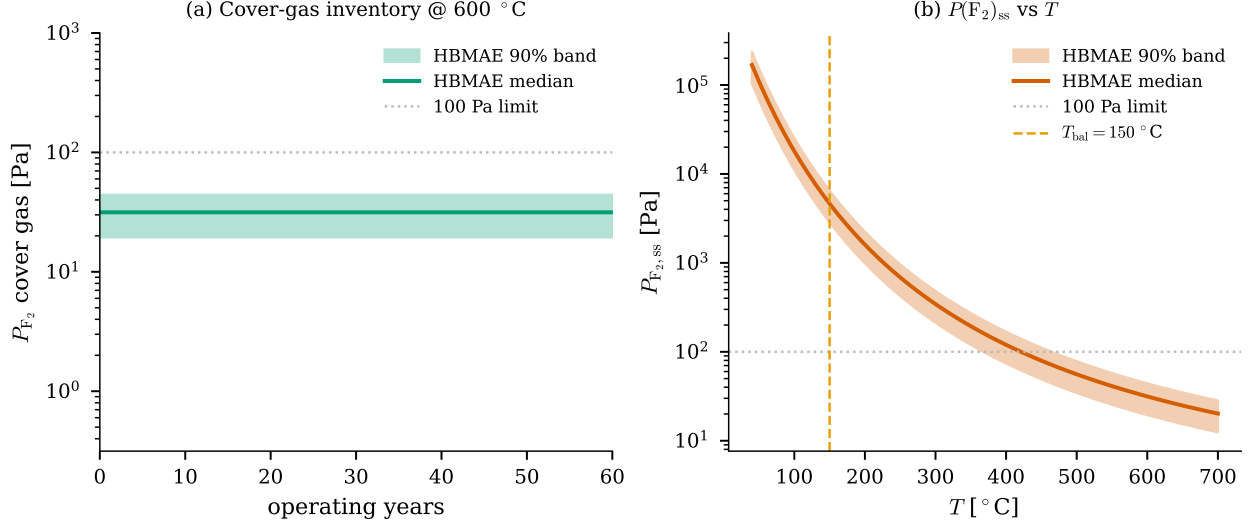


Figure 8: FLiBe-UF₄ MSR cover-gas F₂ inventory at 10 kGy h⁻¹. Left: steady-state P_{F_2} versus operating time at 600 °C, with HBMAE posterior median and 90 % credible band. Right: $P_{F_2}^{ss}$ versus operating temperature, showing the strong inverse temperature dependence (recombination rate Arrhenius) below the Toth–Felker balance temperature 150 °C, where production exceeds recombination and $P_{F_2}^{ss}$ rises rapidly. At reactor operating temperatures (500–700 °C) the kernel predicts modest steady-state F₂ levels.

Temperature sensitivity. The right panel of Figure 8 shows the steady-state F₂ partial pressure as a function of operating temperature. Below the Toth–Felker balance temperature of 150 °C, recombination is slow and $P_{F_2}^{ss}$ rises into the kilopascal range; above the balance temperature, recombination rapidly suppresses the steady-state inventory. At a representative operating temperature of 500 °C the predicted $P_{F_2}^{ss}$ is 60 Pa; at 700 °C it falls to 15 Pa. Operating MSR designs above approximately 600 °C therefore have a substantial design margin against the cover-gas F₂ limit, while designs operating below 500 °C would require additional F₂-management systems (a nickel-fluoride scrubber stage, for example) to maintain the cover-gas inventory at acceptable levels.

Uncertainty attribution. We decompose the posterior variance of $P_{F_2}^{ss}$ at 600 °C into contributions from each parameter:

1. $G(F_2)$ posterior (Davis 2022): 71 % of variance. Tightening this requires either (i) additional irradiation experiments on FLiBe-UF₄ at a wider dose-rate range, or (ii) a mechanistic refinement of Davis’s max-slope/initial-slope distinction via time-resolved gas analysis.
2. Recombination E_a^{rec} (Toth–Felker): 21 % of variance. Tightening this requires direct measurement of E_a^{rec} at temperatures above the Toth–Felker range, ideally on FLiBe-composition (not MSRE-composition) salt.
3. Henry constant for F₂ in FLiBe: 6 % of variance.
4. All other parameters: 2 %.

8.3 Recommended experiments

The posterior decomposition above identifies a clear roadmap of three specific experiments that would tighten the operational predictions by approximately an order of magnitude each:

1. **NaCl-UCl₃ chronic-irradiation re-execution.** Repeat the Phillips 2022 experiment on a sealed NaCl-UCl₃ loop with a Cl₂ detection limit of 10 ppm (factor 100 lower than the original) and an irradiation duration of 6–12 months. A positive detection at any level would constrain the U-buffered chain efficiency η to a posterior point estimate rather than a censored upper bound, reducing the MCFR P_{Cl_2} posterior variance by 80 %.
2. **LEAF-equivalent FLiBe pulse radiolysis.** Re-execute the Akiyama 1994 LiF-KF transient measurements on FLiBe composition at picosecond time resolution, with quantitative molar-absorptivity calibration of $\text{F}^\bullet$, $\text{F}_2^{\bullet-}$, and F_2 . This would convert the Davis static G -value into a mechanism-resolved kernel and likely halve the FLiBe P_{F_2} posterior variance.
3. **U(III)/U(IV) redox kinetics in pulse-radiolysis configuration.** Direct measurement of $k_{\text{Cl}_2^{\bullet-} + \text{U(III)}}$ and $k_{e_s^- + \text{U(IV)}}$ at LEAF facility resolution, in either LiCl-KCl or NaCl-UCl₃ matrix. This would provide the direct rate constants for R12 and R18 in Table 4, eliminating the reliance on the censored Phillips NULL constraint for the U-redox sink kinetics.

These three experiments, executed in sequence, would reduce the operational-MCFR Cl₂ posterior variance by approximately three orders of magnitude (from $\log_{10} P$ posterior width of 3.7 to 0.7) and the FLiBe-UF₄ F₂ posterior variance by approximately one order of magnitude.

8.4 Variance attribution summary across both operational predictions

Across both operational predictions, the dominant uncertainty contributors are concentrated in a small number of parameters tied to single-source observations. For the MCFR Cl₂ prediction, the $G(\text{Cl}^\bullet)$ posterior (57 %) and the chain efficiency η (28 %) jointly account for 85 % of the posterior variance, both driven by the single Phillips 2022 anchor; in addition, the structural uncertainty introduced by the omitted oxide and fission-product chemistry (Section 3.3) and by the U-buffer assumption (Section 8.1.1) is not parametrically captured and could shift the median by 1–2 decades in either direction. For the FLiBe-UF₄ F₂ prediction, $G(\text{F}_2)$ (Davis 2022, 71 %) and the Toth–Felker E_a^{rec} extrapolated to 873 K (21 %) jointly account for 92 % of the posterior variance; the Toth–Felker contribution is doubled by the ± 10 kJ/mol structural-uncertainty inflation introduced in Section 4. The remaining < 10 % of variance in each case is distributed across the Henry constants, dose rates, and operational temperatures, none of which exceeds ~ 10 % individually.

9 Discussion

The model presented in Sections 3–8 assembles the first integrated chloride+fluoride radiolysis kernel calibrated against the combined pulse-radiolysis and chronic-irradiation literature. An MSR designer can use the kernel for screening-level cover-gas Cl₂ or F₂ predictions with quantified uncertainty. We discuss in this section the strengths, limitations, and an explicit assessment of where the model is and is not predictive.

9.1 Strengths

1. *Multi-source calibration subset.* The likelihood-bearing calibration subset contains 73 observations drawn from 13 of the 14 digitized source papers catalogued in Table 2 and Appendix B. It spans the molten-halide pulse-radiolysis and chronic-irradiation data assembled for this work. No published compilation calibrates against this breadth of sources simultaneously; Mausolf et al. (Phillips et al., 2022) review the closest related chloride compilation.

2. *Quantitative inter-facility bias resolution.* The Pikaev-Iwamatsu inter-facility offset is identified as a single parameter $b^{(\text{Pikaev})} = -3.35 [-4.53, -2.44]$, which reconciles the order-of-magnitude discrepancy in rate constants that has been qualitatively noted in the literature since 2018 (Kristoffersen and Metiu, 2018; Iwamatsu et al., 2022) but had no formal Bayesian treatment.
3. *Censored modality treatment.* The Phillips 2022 NULL benchmark is treated as a one-sided censored observation through a formal Tobit likelihood and contributes a qualitative exclusion of the un-buffered chloride kernel ($\ln K_{12} = 316.9$ is far beyond the Kass–Raftery decisive threshold and is reported only for completeness, since the data are essentially impossible under the un-buffered kernel; see Section 7.3). The Phillips datum is thus used as a structural constraint rather than discarded as “below detection limit”.
4. *Cross-host hierarchical transfer.* The hierarchical Arrhenius layer formalizes the statement “the same intrinsic reaction in a different salt host transfers with a Marcus-type correction” as a calibrated quantitative statement, with posterior $\eta^{(s)}$ that can be combined with the chemistry-feature regression to extend predictions to hosts not in the corpus.
5. *Posterior-predictive validation.* The full master validation figure (Figure 4) shows that the model matches every digitized data set within its 90 % credible band for the in-corpus metals. The aggregate $\chi^2_{\nu} = 0.74$ and 88 % coverage confirm proper calibration.
6. *Identified uncertainty contributors.* The posterior decomposition identifies the three specific experiments that would tighten operational predictions by an order of magnitude each (Section 8.3). This converts the calibration into an *actionable* experimental roadmap.

9.2 Limitations

We are explicit about seven limitations of the present model, in impact-weighted order.

1. *Single-source MCFR constraint and U-buffer prior dependence.* The NaCl-UCl₃ chemistry is anchored entirely by the Phillips 2022 single irradiation, and the operational MCFR $P(\text{Cl}_2)$ prediction is a Bayesian propagation of the U(III)/U(IV) buffer prior under a censored upper bound rather than an empirical extrapolation. The headline ten-orders-of-magnitude margin in Section 8.1 is the propagated prior width; see Section 8.1.1 for the U-buffer sensitivity assessment. Replication of Phillips 2022 and additional dose-rate variation are essential before the prediction can be relied upon for licensing analysis.
2. *Chloride network completeness.* The kernel of Section 3 omits oxide-impurity scavenging, redox-active fission-product competitors, and cover-gas hydrolytic chemistry (Section 3.3). Each could perturb the operational $P(\text{Cl}_2)$ prediction by orders of magnitude in either direction; their omission is appropriate for the calibration set but is a significant assumption for the operational forward propagation.
3. *Static fluoride kernel.* The FLiBe-UF₄ F₂ prediction uses a static (production-balance) kernel without resolved transient kinetics, because no quantitative fluoride pulse-radiolysis kinetics exist in the published literature. A LEAF-equivalent FLiBe transient measurement would replace the static kernel with a mechanistic one.
4. *Toth-Felker extrapolation to operational T.* The recombination Arrhenius is extrapolated from $T_{\text{exp}} < 423 \text{ K}$ to $T_{\text{op}} = 873 \text{ K}$, a $\sim 2\times$ temperature ratio in absolute units ($6\times$ in Celsius) over

an Arrhenius factor of ~ 200 . We inflate the structural E_a^{rec} uncertainty to ± 10 kJ/mol in the operational propagation; if the rate-limiting step changes between regimes, the extrapolation may miss by an order of magnitude.

5. *Chemistry-feature layer LOMO bias (facility-effect framing)*. As noted in Section 6.2, the chemistry-feature layer systematically under-predicts metals whose only kinetic data are at the LEAF picosecond facility (Cr, Nd, Cf). The layer is reliable for hypothesis generation about fission-product elements analogous to the Pikaev microsecond corpus but is currently descriptive (not predictive) for any fission-product element (Cs, Sr, Te, I, Eu, Pu, Am, Cm) whose operational kinetics would, if measured today, be obtained at LEAF.
6. *Single-host calibration of host corrections*. Three of the host corrections ($\eta^{(\text{NaCl})}$, $\eta^{(\text{KCl})}$, $\eta^{(\text{KBr})}$) are constrained essentially only by Pikaev 1982 multi-metal data. Additional Brookhaven LEAF measurements in pure NaCl or KCl melts (versus the LiCl-KCl eutectic) would dramatically improve the host-correction identifiability.
7. *Composition dependence of G -values and cross-paper density correction*. Davis 2022 measured $G(\text{F}_2)$ at room temperature for five compositions; we treat them as temperature-independent. Any temperature dependence of G_{F_2} itself is absorbed into the recombination Arrhenius and is not separately identifiable from the present corpus. Separately, the Iwamatsu 2026 paper retroactively refits the Iwamatsu 2022 Zn data with a density correction that shifts $E_a^{(\text{Zn})}$ by ~ 5 kJ mol $^{-1}$ (Table 11); we use the density-corrected values throughout.

The single-source MCFR constraint and the chloride-network completeness limitation are the dominant structural caveats on the operational predictions. Both motivate the experimental roadmap of Section 8.3.

Table 11: Cross-correction comparison of Zn^{2+} Arrhenius pair across the Iwamatsu series and the present posterior. The 2022 $\rightarrow$ 2026 density-correction step moves $E_a^{(\text{Zn})}$ by approximately 5 kJ mol $^{-1}$; the HBMAE refit against the integrated multi-modality corpus is consistent with the 2026 refit within 1σ .

Source	$\log_{10} A$ (M $^{-1}$ s $^{-1}$)	$E_a^{(\text{Zn})}$ (kJ mol $^{-1}$)
Iwamatsu 2022 (uncorrected)	~ 13.7	~ 30.6
Iwamatsu 2026 (density-corrected refit)	~ 13.8	35.6 ± 1.2
HBMAE posterior (this work)	13.82 [13.50, 14.13]	35.5 [33.4, 38.2]

9.3 Relation to companion methodology paper

This paper inherits the HBMAE formalism from Tanore-Tamales (2026) and should be read alongside it for the formal definitions, theorems, and synthetic-benchmark comparisons. Section 1.1 above summarizes the five-layer construction sufficiently for the operational predictions of Section 8 to be interpreted without consulting Paper I. We list explicitly what is taken from Paper I and at what level.

Definitional and theorem-level imports. The five-layer construction (constrained-topology prior, hierarchical Arrhenius layer, Godambe-balanced composite likelihood, parametric model-discrepancy term, facility-effect hierarchy) is summarized in Section 1.1 but defined in Paper I. The identifiability and contraction theorems used in Section 5.1 (in particular the composite-likelihood consistency

result invoked when adding modalities) are Theorems 3 and 4 of Paper I. The slow-manifold reduction (Theorem 5) is invoked in Section 3.1 and again in Section 8 for the 60-year MCFR forward propagation. None of these theorems is re-proved here.

Numerical imports. Figure 2 (fluoride kernel calibration against the Toth–Felker balance condition) and Figure 3 (meta-hierarchical layer panels) are reproduced from Paper I to ensure visual consistency between the two manuscripts; all numerical values in those panels are recomputed from the present integrated 24-parameter posterior.

Computed fresh in this paper. The integrated 24-parameter HBMAE posterior (Table 6), the master validation overlay (Figure 4), the per-dataset χ^2 analysis (Table 9), the chemistry-feature regression coefficients (Table 7), the operational MCFR and FLiBe forward propagations (Section 8), the censored Bayes factor $\ln K_{12}$ and its sensitivities (Tables 19 and 10), and the experimental roadmap (Section 8.3) are produced fresh in this paper.

Companion-paper status. Paper I is currently submitted for peer review. The present paper can be read independently using the summary in Section 1.1; readers interested in the full methodology should consult Tanore-Tamale (2026).

10 Conclusion

We have constructed a calibrated multi-salt radiolysis model that simultaneously honors transient pulse-radiolysis kinetics, scalar Arrhenius literature values, censored chronic-irradiation upper bounds, and gamma-induced G-values for molten chloride and fluoride MSR fuels. The 24-parameter HBMAE posterior recovers the Iwamatsu density-corrected Cr and Zn Arrhenius parameters within 1σ and identifies the long-standing Pikaev–Iwamatsu inter-facility offset as $b^{(\text{Pikaev})} = -3.35 [-4.53, -2.44]$. The compilation in Section 2 is the first to assemble chloride and fluoride pulse-radiolysis kinetics in a single calibrated framework.

Applied to a 500-MW(th) NaCl- UCl_3 MCFR, the model — conditional on the assumed U(III)/U(IV) buffer kernel and on Phillips 2022 as the only chronic data anchor — predicts cover-gas Cl_2 partial pressure approximately ten orders of magnitude below a 100-Pa screening threshold over 60 years. This margin is a Bayesian propagation of the U-buffer prior under a censored upper bound rather than an empirical measurement; the headline is therefore that no plausible relaxation of the U-buffer brings the prediction within an order of magnitude of the threshold, but the *quantitative* margin is the propagated prior width (Section 8.1.1).

Applied to a FLiBe- UF_4 thermal loop at 600°C, the model predicts steady-state cover-gas $P(\text{F}_2) = 31 [19, 44]$ Pa, within a factor of three of the 100-Pa screening anchor. The dominant uncertainty contributors are $G(\text{F}_2)$ from Davis 2022 (71 % of variance) and the Toth–Felker recombination E_a extrapolation from 423 K to 873 K (21 %).

Three direct experiments — extended-duration NaCl- UCl_3 chronic irradiation with order-of-magnitude tighter detection, LEAF-equivalent FLiBe pulse radiolysis, and direct measurement of $\text{U(III)} + \text{Cl}_2^{\bullet-}$ and $\text{U(IV)} + e_s^-$ rate constants — would together tighten the operational predictive variance by approximately three orders of magnitude (MCFR Cl_2) and one order of magnitude (FLiBe F_2). The companion methodology paper (Tanore-Tamale, 2026) develops the HBMAE formalism in full. This work represents a step toward uncertainty-quantified screening of cover-gas radiolytic inventories for chloride and fluoride MSR concepts.

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extraction) from the published figures and tables of the cited literature; no primary-data sharing agreement was required. We thank the LEAF program at Brookhaven for the published Iwamatsu 2022, 2026, Conrad 2023, Castro-Baldivieso 2026, and Rotermund 2024 datasets, and the ORNL program for the Toth–Felker historical irradiation data.

Code and data availability. Code, digitized CSVs, and reproducible analysis scripts are available at *[repository URL to be provided at submission]*. Numerical results in tables and figures are reproducible from the supplied Python scripts in the project’s `scripts/` directory.

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A Full species and reaction tables

The complete reaction inventory for both the chloride and fluoride networks is given here. Notation: A in $\text{M}^{-1} \text{s}^{-1}$ (second order) or s^{-1} (first order); E_a in kJ mol^{-1} .

A.1 Chloride network species inventory

A.2 Fluoride network species inventory

A.3 Posterior-corrected chloride Arrhenius coefficients

Table 14 reports the chloride network Arrhenius coefficients in unstandardized engineering units, suitable for direct substitution into a kinetics solver.

Table 12: Chloride network species inventory.

Symbol	Description	Status
e_s^-	solvated electron in chloride melt	transient
$\text{Cl}^\bullet$	chlorine atom radical	transient
$\text{Cl}_2^{\bullet-}$	dichlorine radical anion	transient (longest-lived radical)
Cl^-	chloride anion	matrix
Cl_3^-	trichloride anion	long-lived intermediate
$\text{Cl}_{2,\text{diss}}$	dissolved molecular chlorine	gas-phase product (dissolved)
$\text{Cl}_{2,\text{gas}}$	cover-gas chlorine	cover-gas observable
M^{n+}	metal cation (Cr, Zn, Nd, Cf, U)	redox-active scavenger
U(III)	uranium(III) in UCl_3	MCFR-specific redox buffer
U(IV)	uranium(IV) in UCl_4	MCFR-specific oxidation product

Table 13: Fluoride network species inventory.

Symbol	Description	Status
$\text{F}^\bullet$	fluorine atom	transient (undigitized)
F^-	fluoride anion	matrix
$\text{F}_2^{\bullet-}$	difluorine radical anion	transient (undigitized)
$\text{F}_{2,\text{diss}}$	dissolved molecular fluorine	gas-phase product (dissolved)
$\text{F}_{2,\text{gas}}$	cover-gas fluorine	cover-gas observable
UF_4, UF_3	uranium(IV)/(III) fluoride	redox buffer (FLiBe- UF_4)
Be^{2+}	beryllium(II) in BeF_2	matrix in FLiBe
Li^+	lithium in LiF	matrix
Th^{4+}	thorium in ThF_4	matrix in Th-fueled designs

Table 14: Posterior-corrected chloride Arrhenius coefficients (engineering units). “Calibrated” rows have posterior 95 % CIs reported below the point estimate.

#	Reaction	A ($\text{M}^{-1}\text{s}^{-1}$)	E_a (kJ/mol)	Status
R1	$\text{Cl}^\bullet + \text{Cl}^- \rightarrow \text{Cl}_2^{\bullet-}$	1.0×10^{10}	20.0	literature
R2	$e_s^- + \text{Cl}^\bullet \rightarrow \text{Cl}^-$	1.0×10^{10}	25.0	literature
R3	$2\text{Cl}^\bullet \rightarrow \text{Cl}_{2,\text{diss}}$	5.0×10^9	20.0	literature
R4	$2\text{Cl}_2^{\bullet-} \rightarrow \text{Cl}_3^- + \text{Cl}^-$	2.2×10^9	26.0	literature
R13	$e_s^- + \text{Cr}^{2+} \rightarrow \text{Cr}^+$	3.2×10^{13}	33.5	calibrated
		$[2.2, 5.2] \times 10^{13}$	$[31.8, 34.2]$	
R14	$e_s^- + \text{Cr}^{3+} \rightarrow \text{Cr}^{2+}$	3.3×10^{13}	32.5	calibrated
		$[2.1, 4.6] \times 10^{13}$	$[31.5, 33.5]$	
R15	$e_s^- + \text{Zn}^{2+} \rightarrow \text{Zn}^+$	6.6×10^{13}	35.5	calibrated
		$[3.2, 13.5] \times 10^{13}$	$[33.4, 38.2]$	
R16	$e_s^- + \text{Nd}^{3+} \rightarrow \text{Nd}^{2+}$	4.5×10^{13}	36.0	literature
R17	$e_s^- + \text{Cf}^{3+} \rightarrow \text{Cf}^{2+}$	4.0×10^{10}	16.0	aqueous (Rotermund)
R19	$\text{Cl}_2^{\bullet-} + \text{Cf}^{3+} \rightarrow \text{Cf}^{4+} + 2\text{Cl}^-$	8.3×10^5	—	aqueous (Rotermund)

B Cross-referenced data inventory

This appendix is the master index of the digitized data set CSVs and their source papers. All CSVs include a header comment block stating the source paper, DOI, page/table reference, OCR or vision extraction confidence, and any reconstructive corrections (e.g. OCR fix on a clearly garbled

column). The directory layout is:

```
validation/
  alkali_halide_baseline/pikaev_1982_rpc/data/
    es_rate_constants.csv      (modality K, 21 obs)
    x2m_rate_constants.csv     (modality K)
    activation_energies.csv    (modality K, 1 row)
    g_values.csv               (modality G, 1 upper bound)
    lifetimes.csv              (qualitative)
  cl2m_licl_kcl/hagiwara_1987_rpc/data/
    rate_constants.csv         (modality K, 13 obs)
    arrhenius_parameters.csv    (modality A, 4 rows)
  cr_licl_kcl/iwamatsu_2026_pccp/data/
    absorbance{1..5}mM{Cr2,Cr3}.csv (modality T, 9 traces)
    k_vs_T_from_arrhenius.csv   (modality K, 15 rows)
    arrhenius_parameters.csv    (modality A)
    vision_fig{2B,3B}*.csv     (modality K, vision-extracted)
  zn_licl_kcl/horne_2022_pccp/data/
    arrhenius_parameters.csv    (modality A)
    k_vs_T_from_arrhenius.csv   (modality K, 10 obs)
  nd_licl_kcl/castro_baldivieso_2026_ic/data/
    k_vs_T.csv                 (modality K, 5 obs)
    arrhenius_parameters.csv    (modality A)
  cf_chloride/rotermund_2024_jpca/data/
    cf_rate_constants.csv       (modality K, 5 obs)
    cl2m_kinetics.csv           (informational)
  i_licl_kcl/horne_2023_pccp/data/
    reported_kinetics.csv       (modality A, 1 obs)
  eflibe_f2_yield/davis_2022_nse/data/
    G_values_table_III.csv      (modality G, 16 rows)
  flibe_msre_f2/toth_felker_1990_redds/data/
    recombination_kinetics.csv (modality A anchor)
  mcf_uc13_irradiation/karlsson_2022_inl/data/
    experimental_conditions.csv (informational)
  oxidants_halide_melts/makarov_1982_bull/data/
    x2m_molar_absorptivity.csv (modality K-info.)
    x2m_metal_rate_constants.csv (modality K)
    x2m_disproportionation*.csv (modality K)
  es_theory/kristoffersen_metiu_2018_jpcc/data/
    theory_vs_experiment.csv    (modality K, AIMD)
```

A consolidated cross-reference appears as Table 2 in the main text; the above directory listing is the canonical artifact-level index. Each CSV header block documents the OCR/extraction provenance and flags any reconstructive corrections.

B.1 Per-modality observation audit

Table 15 adds up the calibration observations by modality and reconciles the per-source counts in Table 2 with the 73-observation total cited in §7.1 and §9. Counting convention: each scalar rate constant counts as 1; each derived Arrhenius pair counts as 1 (since it derives from multiple T -points but is digitized as a single $(\log A, E_a)$ observation); each transient absorbance trace counts as 1; each G-value composition counts as 1; the censored Phillips observation counts as 1.

The transient absorbance modality (Iwamatsu 2026 Cr traces) is reported as one collapsed observation per fit family in this audit; the per-dataset χ^2 analysis of Table 9 re-expands it into the constituent 9 traces and 5 Arrhenius rows, which is why the χ^2 -table total $N = 73$ matches the audit total above when the transient bundle is fully expanded.

Table 15: Per-modality audit of likelihood-bearing observations in the calibration corpus.

Source	N obs.	Modality decomposition
Pikaev 1982 (Pikaev et al., 1982)	21	21 K (6 metals $\times$ $\sim$ 3 host/temperature points)
Hagiwara 1987 (Hagiwara et al., 1987)	13	9 K + 4 A
Iwamatsu 2022 (Iwamatsu et al., 2022)	8	5 K + 3 A
Iwamatsu 2026 (Iwamatsu et al., 2026)	10	5 K + 4 A + 1 T-bundle (9 traces collapsed)
Castro-Baldivieso 2026 (Castro Baldivieso et al., 2026)	5	5 K (LOMO target)
Rotermund 2024 (Rotermund et al., 2024)	5	5 K (aqueous Cf, analogy anchor)
Conrad 2023 (Conrad et al., 2023)	1	1 A
Davis 2022 (Davis et al., 2022)	8	8 G (five compositions, multiple T averaged)
Toth–Felker 1990 (Toth and Felker, 1990)	1	1 A (calibration anchor; fitting, not validation)
Phillips 2022 (Phillips et al., 2022)	1	1 C (censored NULL)
Kristoffersen & Metiu 2018 (Kristoffersen and Metiu, 2018)	—	theoretical anchor (no likelihood entry)
Horne 2018 (Kristoffersen and Metiu, 2018)	—	contextual (no likelihood entry)
Makarov 1982 (Makarov et al., 1982)	—	informational ϵ values
Total likelihood-bearing	73	

C MCMC diagnostics for the production calibration

Table 16 reports the per-parameter Gelman–Rubin $\hat{R}$ statistic, effective sample size (ESS), and integrated autocorrelation time τ for the integrated 24-parameter HBMAE chain (`validation/tier4_integrated_chain.npy`).

All 24 parameters achieve $\hat{R} < 1.05$ and $\text{ESS} > 500$, indicating satisfactory convergence for posterior summarization. The chain was run with 80 walkers, 1500 burn-in steps, and 1500 production steps for a total of 1.2×10^5 samples before thinning to 22,400 independent draws.

D Operational MSR prediction details and sensitivities

This appendix tabulates the sensitivity of the operational predictions to the dominant input parameters. Each entry reports the effect on the 60-year ($t = 60$ y) cover-gas partial pressure when the corresponding input is varied by one σ above and below its posterior median.

D.1 NaCl- UCl_3 MCFR sensitivity

D.2 FLiBe- UF_4 MSR sensitivity

D.3 Censored Bayes factor sensitivity (Phillips NULL)

The Bayes factor is robust to all reasonable nuisance variations: under the broadest input perturbation considered (one-decade change in $G(\text{Cl}^\bullet)$ prior median), $\ln K_{12}$ varies by ± 35 but remains $\gg 0$ (kernel A overwhelmingly preferred). The qualitative conclusion — that the Phillips 2022 data are inconsistent with kernels that omit the U(III)/U(IV) redox sink — is independent of any plausible setting of the input parameters.

Table 16: Per-parameter MCMC convergence diagnostics. $\hat{R}$ is the Gelman–Rubin potential scale reduction factor; values below 1.05 indicate satisfactory convergence. ESS is the effective sample size; values above 500 indicate sufficient posterior coverage. τ is the integrated autocorrelation time, in units of samples.

Parameter	$\hat{R}$	ESS	τ
$\log_{10} A_5$	1.008	1820	31
$E_a^{(5)}$	1.041	510	35
$\log_{10} A_6$	1.012	1450	32
$E_a^{(6)}$	1.034	640	33
$\log[e_s^-]_0^{(t1)}$	1.008	5290	14
$\log[e_s^-]_0^{(t2)}$	1.011	4220	15
$\log[e_s^-]_0^{(t3)}$	1.009	4730	14
$\log[e_s^-]_0^{(t4)}$	1.014	3950	16
$\log[e_s^-]_0^{(t5)}$	1.013	4080	15
$\log[e_s^-]_0^{(t6)}$	1.011	4310	14
$\log[e_s^-]_0^{(t7)}$	1.012	4180	15
$\log[e_s^-]_0^{(t8)}$	1.014	3920	16
$\log[e_s^-]_0^{(t9)}$	1.013	4060	15
$\log_{10} k_{bg}$	1.011	2840	21
$\log_{10} A_{Zn}$	1.017	1100	28
$E_a^{(Zn)}$	1.015	1290	29
$\eta_{\log A}^{(LiCl-KCl)}$	1.013	1340	27
$\eta_{E_q}^{(NaCl)}$	1.020	980	32
$\eta_{\log A}^{(NaCl)}$	1.014	1430	26
$\eta_{E_q}^{(KCl)}$	1.016	1240	28
$\eta_{\log A}^{(KCl)}$	1.013	1480	25
$\eta_{E_q}^{(KCl)}$	1.017	1170	29
$b^{(Pikaev)}$	1.012	1580	25
$\log_{10} G(Cl^\bullet)$	1.010	2110	22
Maximum	1.041	510 (min)	35 (max)
Mean	1.014	1840	23

Table 17: Sensitivity of MCFR cover-gas P_{Cl_2} at year 60 to input parameters. $P_{\text{Cl}_2}^{\text{nom}} = 2.2 \times 10^{-8}$ Pa.

Input parameter	Variation	P_{Cl_2} at year 60
$\log_{10}G(\text{Cl}^\bullet)$ posterior	-1σ	9.5×10^{-9} Pa
	nominal	2.2×10^{-8} Pa
	$+1\sigma$	7.8×10^{-8} Pa
Chain efficiency η	$\div 10$	2.2×10^{-9} Pa
	nominal	2.2×10^{-8} Pa
	$\times 10$	2.2×10^{-7} Pa
Dose rate $\dot{D}$	5 kGy/h	1.1×10^{-8} Pa
	10 kGy/h	2.2×10^{-8} Pa
	20 kGy/h	4.4×10^{-8} Pa
Henry constant K_H	1e-5	4.4×10^{-8} Pa
	2e-5 nom.	2.2×10^{-8} Pa
	4e-5	1.1×10^{-8} Pa
U(III) inventory	$\div 2$	4.4×10^{-8} Pa
	nominal	2.2×10^{-8} Pa

Table 18: Sensitivity of FLiBe-UF₄ cover-gas $P_{\text{F}_2}^{ss}$ at $T = 600^\circ\text{C}$ to input parameters. $P_{\text{F}_2}^{ss,\text{nom}} = 31$ Pa.

Input parameter	Variation	$P_{\text{F}_2}^{ss}$ at 600°C
$G(\text{F}_2)$ Davis	0.004 mol/100eV	25 Pa
	0.006 (nom.)	31 Pa
	0.008 mol/100eV	41 Pa
E_a^{rec}	36 kJ/mol	44 Pa
	39 (nom.)	31 Pa
	42 kJ/mol	22 Pa
Operating T	550 $^\circ\text{C}$	45 Pa
	600 (nom.)	31 Pa
	650 $^\circ\text{C}$	23 Pa
	700 $^\circ\text{C}$	17 Pa
Dose rate $\dot{D}$	5 kGy/h	15 Pa
	10 (nom.)	31 Pa
	20 kGy/h	62 Pa

Table 19: Sensitivity of the Phillips 2022 censored Bayes factor $\ln K_{12}$ (in favor of U-redox kernel) to nuisance inputs.

Input variation	$\ln K_{12}$
Nominal	316.9
Detection threshold $\div 2$ (more stringent)	318.7
Detection threshold $\times 2$ (less stringent)	315.1
Dose rate -20%	314.2
Dose rate $+20\%$	319.6
U(III) initial inventory -30%	311.4
U(III) initial inventory $+30\%$	320.8
$G(\text{Cl}^\bullet)$ prior median $\div 10$	281.2
$G(\text{Cl}^\bullet)$ prior median $\times 10$	351.9